

# Crossing State Lines: Licensure Compacts, Provider Mobility, and Mental Health

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## Abstract

Despite the high prevalence of mental illness, many Americans live in areas with insufficient mental health professionals. Interstate licensure compacts aim to address provider shortages by reducing barriers to practicing across state lines. Leveraging staggered adoption of the Interstate Medical Licensure Compact (IMLC) and Psychology Interjurisdictional Compact (PSYPACT), I examine effects on multi-state licensure, self-reported mental health, SSRI prescribing, and suicide rates. I find that compact participation increases multi-state licensure among eligible providers but find no consistent evidence of improved population mental health outcomes, suggesting that reducing administrative barriers to interstate practice alone may be insufficient to address the mental health crisis.

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# 1 Introduction

Nearly one in four adults in the United States struggle with mental illness. Mental illnesses encompass a wide range of conditions that differ in severity, from mild to moderate to severe (NIMH). Despite the high prevalence, most Americans live in an area without enough mental health professionals to meet the population’s treatment needs. In fact, only half of all adults with any mental illness received some type of treatment in 2023 (SAMHSA, 2024).

One way in which policy makers have begun to try to alleviate this problem is by easing healthcare worker mobility through the use of interstate licensure compacts that streamline the licensing process for healthcare workers who wish to practice in multiple states. Because medical practice is regulated individually by each state, physicians must otherwise obtain a separate license for each state in which they wish to practice. The lack of a unified national system creates administrative burdens and limits physician mobility. By streamlining the process of obtaining licenses to practice in multiple states, interstate compacts have the potential to allow labor to flow freely across states.

At the same time, due to improvements in technology and the COVID-19 pandemic, telehealth care has expanded rapidly. This shift towards remote healthcare could incentivize providers to expand their practice across state lines, increasing the need for interstate licensure compacts.

Recent research shows that interstate licensure compacts increase out-of-state physician practices and the odds that a mental health facility offers telehealth services (Deyo et al., 2023; McBain et al., 2023). These findings suggest that compact participation can facilitate the provision of care across state lines and create opportunities for expanded access to care. However, whether these changes ultimately translate into improved health outcomes remains an important question. There is no existing empirical research examining the downstream effects of these compacts on patient health, leaving a gap in our understanding of their broader impact. Studying the impacts of these compacts on health outcomes is crucial because it helps to determine whether the increased availability of healthcare providers and services actually leads to better patient health. Furthermore, evaluating health outcomes sheds light on whether vulnerable populations, such as those in rural or underserved areas, experience meaningful benefits, or if other factors like treatment quality need further attention.

In this paper, I fill the gap in the literature by examining whether interstate licensure compacts ultimately affect population mental health outcomes. I utilize the staggered adoption of the Interstate Medical Licensure Compact and the Psychology Interjurisdictional Compact to first examine whether compact participation increases multi-state licensure among eligible providers and then estimate effects on self-reported mental health, antidepressant prescribing rates, and suicide rates. I find that compact participation increases multi-state licensure among eligible providers. However, I find little evidence that this provider response translates into improvements in population mental health. Compact participation has no significant effect on average self-reported poor mental health days, and in some cases, slightly increases them, particularly among individuals previously diagnosed with depression. I also find no substantial impact on the incidence of major depressive disorder or SSRI prescribing rates. While PSYPACT participation is associated with a statistically significant decline in state suicide rates, the magnitude of this effect is small. Overall, my findings suggest that reducing barriers to interstate practice can increase multi-state licensure, but that expanded licensure does not translate into better mental health outcomes.

## 2 Background

### 2.1 Mental Health Treatment

Mental health care involves a range of professionals, each with specific roles in diagnosis and treatment. Primary care physicians (PCPs) are often the first point of contact for individuals seeking help. They can conduct initial screenings, provide basic diagnoses, prescribe common mental health medications like antidepressants, and refer patients to mental health specialists when necessary. Psychologists specialize in providing therapy. They are trained to assess mental health conditions and offer a variety of therapeutic treatment approaches, but they generally cannot prescribe medication. Psychiatrists are medical doctors who can diagnose mental illnesses, prescribe and manage medications, and may also offer therapy.

Treatment options for mental health care typically fall into two main categories: therapy and medication. Therapy includes structured sessions that help individuals address emotional challenges, develop coping strategies, and change unhelpful thinking or behavior patterns. A wide range of psychotherapeutic interventions have demonstrated effectiveness in treating mental health disorders (Barth et al., 2013; Roy-Byrne et al., 2005). Medications, such as antidepressants, are often most effective for reducing symptoms quickly and preventing relapse, particularly in cases of severe depression (Eells, 1999).

These treatments can be delivered in different formats to suit a patient’s preferences and circumstances. In-person treatment involves face-to-face interaction at a clinic or office, while telehealth services, conducted through phone or video, allow for remote access to care. Telepsychiatry has proven to be a reliable and effective option, producing outcomes comparable to in-person care across a range of conditions including depression, anxiety, PTSD, and substance use disorders (Sharma and Devan, 2023). It has also demonstrated high levels of patient satisfaction, increased treatment adherence, and improved access (Ruskin et al., 2004; Urness et al., 2006).

### 2.2 Professional Licensing

States began licensing physicians through state medical boards in the late 19th century. In 1912, the Federation of State Medical Boards (FSMB) was founded, and by 1979 all 50 states issued medical licenses in accordance with the Federation Licensing Examination (FLEX). While the adoption of this examination offers uniformity among licensing requirements, medical practice and licensure are still regulated by individual states.

The practice of telemedicine is also regulated by each state individually. Thus, for the most part, physicians who wish to practice in multiple states, even virtually, are required to be licensed to practice in each state. As telemedicine grows in acceptance and popularity, these licensure regulations are a significant barrier to wider expansion. For this reason, advocates for telemedicine have been calling for a cooperative, interstate approach to medical licensure since the late 2000s (Adams et al., 2018; Cwiek et al., 2007).

Recognizing the increasing desire of physicians to practice in multiple states as a result of telemedicine, the Interstate Medical Licensure Compact (IMLC) was introduced by a group of state medical board executives with assistance from the Federation of State Medical Boards in the fall of 2014. With the first states adopting legislation in 2015, the compact became operational in April 2017. The IMLC aims to increase access to healthcare by extending the reach of physicians and leveraging technology, such as telemedicine. While licensure is still regulated by each state, the IMLC streamlines the licensing process by allowing eligible physicians to complete a single

application within the compact in order to receive licenses from each state in which they intend to practice. Around 80 percent of U.S. physicians meet the criteria for the compact, including psychiatrists (IMLC Commission).

More recently, the Psychology Interjurisdictional Compact (PSYPACT) was developed to facilitate the practice of telepsychology across state lines. Similar to the IMLC, PSYPACT allows psychologists to file a single application to practice telepsychology across PSYPACT participating states. The compact became effective in 2020 for early adopters of PSYPACT legislation. As of 2023, 38 states have effective PSYPACT legislation (PSYPACT Commission).

## 2.3 Compacts and Practitioner Incentives

One major goal of interstate licensure compacts is to increase the availability of medical services and leverage the use of telemedicine. Interstate compacts attempt to increase access to care by increasing the labor mobility of healthcare workers. By reducing the administrative costs associated with obtaining authorization to practice in additional states, these compacts create an opportunity for providers to expand their practices across state lines. However, streamlined licensure does not necessarily imply that eligible providers will obtain additional licenses or provide care to patients in additional states. Healthcare professionals practicing in participating states have the option to become licensed in multiple participating states or continue to solely be licensed in their original practice state. If a professional chooses not to apply for multi-state licensure through the compact, then he must continue to practice only in the state(s) in which he is already licensed. If a professional chooses to take advantage of the compact's streamlining process, however, he now has the option to continue practicing in his original practice state in addition to practicing in other participating states via commuting or telemedicine or to migrate his practice to another participating state.

Conditional on providers obtaining authorization to practice in additional states, each of these choices introduces tradeoffs. Among those opening new out-of-state practices, changes in the geographic composition of patients are possible. A healthcare professional who begins practicing in multiple states may choose to increase his working hours to see patients in new practice states in addition to those he was already treating in his original practice state. If he chooses not to change his working hours, a professional can instead decide to shorten visit lengths in order to see more patients or to maintain visit length while changing the in/out-of-state patient mix. Shortening visit lengths could allow a professional to continue caring for his current patients within his original practice state in addition to new patients in new practice states. Keeping the visit length the same requires the professional to stop providing care to some patients from the original practice state in order to have time to treat those in new practice states.

Financial considerations may further influence these choices. In many cases, telehealth visits are reimbursed at lower rates than in-person care, which could discourage some providers from adopting or expanding telehealth services. However, several states have enacted payment parity laws that require insurers to reimburse telehealth services at the same rate as in-person visits, reducing this disincentive. Medicaid reimbursement policies also vary significantly by state, with some offering higher rates than others. These differences in reimbursement structures can affect where and how providers choose to practice, ultimately shaping the composition and distribution of the patient populations they serve across state lines.

The effect of compact participation on mental health outcomes, therefore, depends both on whether providers respond to streamlined licensure by obtaining authorization to practice in additional states and on how they subsequently allocate care across patients and states. If mental health professionals do not take advantage of streamlined licensure, there is less scope for compact

participation to affect access to care or mental health outcomes. On the other hand, if a professional chooses to migrate his practice exclusively to a new state, then mental health could improve among patients in the new practice state while declining among those in the old practice state.

Because the change in patient geographic composition is ambiguous for those who choose to continue practicing in their original practice state in addition to practicing in other participating states via commuting or telemedicine, the expected change in mental health outcomes is also uncertain. If a professional chooses to increase his patient load by treating new out-of-state patients in addition to his established in-state patients, mental health outcomes in new practice states might see improvements while those in the old practice state may remain unchanged. If, however, a professional keeps his patient load constant, then in order to treat patients in new-practice states, he must terminate treatment of some established patients. In this case, there will be fewer patients being treated in the original practice state and more individuals being treated in the new practice states so that health outcomes decline and improve in the respective states.

There are additional nuances that may impact the magnitude and direction of changes in mental health outcomes that result from these changes in patient composition. It is possible that increasing the patient load through either increases in working hours or decreases in visit length leads to declines in the quality of care a professional provides (Geraghty et al., 2007; Hutton and Gunn, 2007). This would then lead to declines in outcomes for original patients which could negate any improvements in aggregate outcomes. On the other hand, professionals may be choosing to treat additional patients on a basis of severity. In which case, the benefits may outweigh the costs and lead to overall improvements in mental health. Thus, it is unclear whether the opportunity to expand access to care created by interstate licensure compacts will lead to improvements in overall health.

Empirical evidence on how providers actually respond is mixed. While it is possible that compact participation motivates professionals to migrate their practices, evidence has shown that this may not be occurring (DePasquale and Stange, 2016; Fannin, 2020). There is some evidence, however, that the Nurse Licensure Compact (NLC) led to a significant increase in nurse mobility (Shakya et al., 2022).

Beyond traditional labor market measures, other studies examine how compact participation affects where and how providers deliver care. There is evidence that IMLC participation is associated with increases in out-of-state practices among physicians whose primary state participates (Deyo et al., 2023).<sup>1</sup> In contrast, Goetz (2026) finds that PSYPACT increases the number of psychologists listing services in participating states, but finds little evidence that this increase reflect entry by out-of-state psychologists. McBain et al. (2023) also find that both IMLC and PSYPACT participation correspond with increased provision of telehealth services. In the nursing sector, higher out-of-state telemedicine use is associated with NLC participation, though the difference is not statistically significant (Norris and Nandy, 2023).

Taken together, these findings suggest that while licensure compacts expand professional options in theory, actual changes in practice patterns vary by profession and over time. However, whether streamlined licensing leads eligible providers to obtain authorization to practice in additional states remains an important part of understanding the policy’s effect on access to care. Even when providers expand their interstate practice, the effects of compact participation on patient composition and care delivery are difficult to observe directly and likely depend on provider-specific and contextual factors. As a result, the net impact of interstate licensure compacts on popula-

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<sup>1</sup>Out-of-state practices refer to practice locations outside of a physician’s state of initial practice, or resident state. These are practices added in addition to resident state practice locations.

tion health outcomes remains uncertain, and an empirical investigation will help shed light on this relationship.

## **3 Data**

### **3.1 Provider Licensure Data**

To examine whether interstate licensure compact participation increases multi-state licensure among eligible providers, I construct a provider-year panel using annual National Plan and Provider Enumeration System (NPES) files from 2017–2023. These files report the states in which each provider holds an active professional license, allowing me to measure the number of states in which each provider is licensed over time. I restrict the sample to providers initially observed with a license in only one state and define this state as the provider’s origin state of licensure. Providers are required to remain licensed in their origin state throughout the sample period, ensuring that compact exposure is consistently defined and avoiding changes in treatment status due to provider relocation. Finally, I use Centers for Medicare and Medicaid Services Doctor and Clinician National Downloadable Files to restrict by provider specialty.

I use these data to construct several measures of multi-state licensure. I define an indicator equal to one if a provider holds active licenses in more than one state in a given year. I also measure the number of states in which each provider holds an active license and construct an ordered measure that categorizes providers according to the number of states in which they are licensed.

### **3.2 Mental Health Outcomes**

Mental health can be measured using a range of outcomes that reflect different levels of severity and types of experiences. In this analysis, I focus on four key indicators: the number of poor mental health days reported in the past 30 days, the incidence of major depressive disorder (MDD), suicide rates, and the prescribing rate of selective serotonin reuptake inhibitors (SSRIs).

By studying an outcome such as the count of poor mental health days one reports having in the last month, I capture the effects of interstate compacts on individuals experiencing even very mild symptoms of mental illness. In the same way, studying the effects on suicide rates allows me to capture the impact of compact participation on the most severe cases. Suicide is often linked to untreated or severe mental illness, making it a critical indicator of overall mental health in a population (Fu et al., 2021; Schmutte et al., 2021; Yeh et al., 2019). Finally, measuring the incidence of MDD and the SSRI prescribing rate provide insights into the effects on the community between these two extremes.

While decreases in poor mental health days, major depressive disorder, and suicide rate all indicate improvements in mental health, it is unclear in which direction the SSRI prescribing rate will move. An increase in the SSRI prescribing rate could indicate that more people are in need of medication than before, but it could also imply that more people are able to access care that was needed prior to the policy change. On the other hand, a decrease in prescribing rate may suggest a decline in the need for medication as a result of increased access to psychotherapy.

#### **3.2.1 Individual-level Data**

Data on the number of poor mental health days for individuals come from the Behavioral Risk Factor Surveillance System (BRFSS) for 2011–2023. BRFSS is an annual national survey of U.S.

residents’ health-related behaviors, conditions, and service usage. Individuals are asked “thinking about your mental health, which includes stress, depression, and problems with emotions, for how many days during the past 30 days was your mental health not good?” Responses to this question provide a measure of the count of poor mental health days in the last month. In addition to this count variable, I create two binary variables to approximate a measure of the incidence of major depressive disorder (MDD). According to the Diagnostic and Statistical Manual of Mental Disorders, 5th Edition, Text Revision (DSM-5-TR), to be diagnosed with a major depressive episode a patient must have five or more depressive symptoms for two weeks continuously. Thus, my first measure of MDD includes all individuals who report having at least 14 poor mental health days in the last 30. Since the clinical definition requires symptoms to persist for at least two consecutive weeks, however, this measure most likely overestimates the incidence of MDD. Therefore, I include a secondary measure of MDD defined as individuals who report having at least 21 poor mental health days in the last 30.

I run my main analyses on both the full sample and the sample of individuals who report having ever been diagnosed with depression. It is reasonable to think that individuals who have been diagnosed with depression are likely to be more responsive to changes in access to care than others. Therefore, the effects of compact participation may differ for this population. The variable used to determine a depression diagnosis includes all surveyed individuals who report having ever been diagnosed with depression. Because BRFSS does not have a longitudinal survey design, I am, therefore, unable to differentiate between newly diagnosed individuals and those who previously had a diagnosis. Thus, I opt to use a diagnosis as a sample restriction rather than as an outcome in my analysis. Columns 1 and 2 of Table 1 show summary statistics for the entire BRFSS sample and the sample restricted to those who have ever been diagnosed with depression, respectively.

### 3.2.2 State-level Data

State suicide rates come from the Multiple Cause of Death files from the Centers for Disease Control and Prevention (CDC) from 2011-2023. These data are based on death certificates of U.S. residents. The data contains the number of deaths, crude death rates, and age-adjusted death rates by cause of death for each year. I collect age-adjusted death rates by state for causes of death categorized under “Intentional self-harm” (ICD-10 codes X60-X84). This includes deaths for which intentional self-harm was recorded as the underlying cause of death or as one of multiple causes of death.

To measure the antidepressant prescription rate, I obtain data from the Centers for Medicare and Medicaid Services Medicare Part D Prescriber files from 2013-2022. This dataset contains information on prescription drugs prescribed by health care providers and paid for under the Medicare Part D Prescription Drug Program. For each drug, the dataset includes the total number of prescriptions that were dispensed, which includes original prescriptions and any refills. To obtain an antidepressant prescription rate for each state, I aggregate the total number of prescriptions of five selective serotonin reuptake inhibitors (SSRIs) and divide this by the sum of all prescriptions dispensed in a state.<sup>2</sup> Descriptive statistics of state-level outcomes are presented in column 3 of Table 1.

All datasets, excluding prescription data, are nationally representative. Because I was unable to obtain the entire universe of prescription data, I am limited to that for the Medicare population. Therefore, analyses utilizing prescription data are not necessarily generalizable and should be

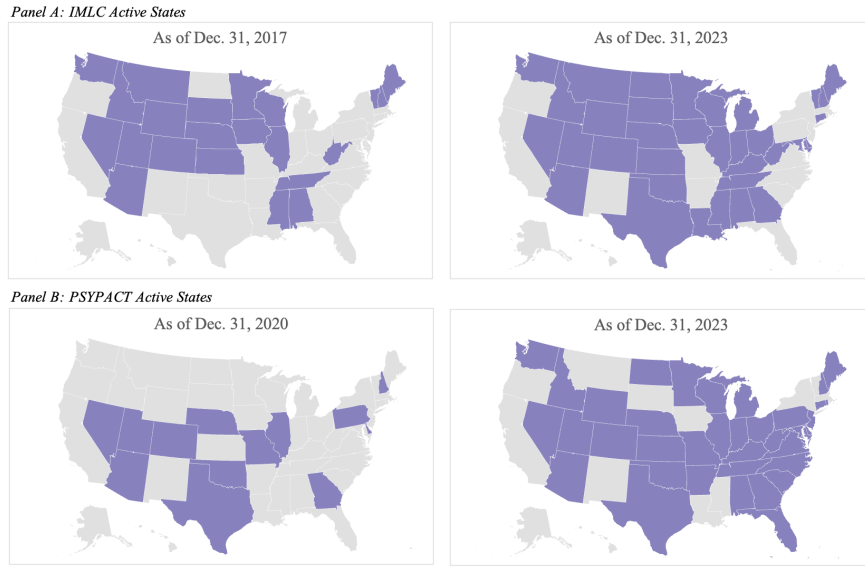
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<sup>2</sup>The five SSRIs included in calculating the prescribing rate are: Fluoxetine (Prozac), Paroxetine (Paxil, Pexeva), Sertraline (Zoloft), Citalopram (Celexa), Escitalopram (Lexapro)

interpreted as such.

### 3.3 Interstate Licensure Compacts

Information on compact participation comes directly from the IMLC and PSYPACT. In my analysis, I define the active date as the start of participation in both the IMLC and PSYPACT rather than the introduction or passage date. This is because a provider cannot obtain an interstate license through the compact until legislation is enacted and active in his state. The IMLC became operational in April 2017, and PSYPACT became operational in July 2020. States' adoption of the IMLC and PSYPACT are staggered following the date each compact became operational. Within my analysis period, 34 states and the District of Columbia joined the IMLC, and 37 states and the District of Columbia joined PSYPACT. Figure 1 shows state participation in the IMLC and PSYPACT over time.<sup>3</sup>



**Figure 1:** Compact Participation over Time

### 3.4 Additional Variables

State-level demographic controls come from the American Community Survey (ACS) for 2011-2023. I include the percent of the population that is Black as well as the percent Hispanic. In addition, variables are included for the unemployment rate and the percentage of the population that is not insured. Because compact participation is associated with an increased probability of telehealth care being offered by a mental health facility, those with access to internet have the potential to see a greater impact than those without access. Therefore, I also control for the percentage of the population without access to internet within each state. For the years 2011 and 2012, I construct the percent of the population without internet by subtracting the percent of individuals who live in a household with internet use from 100. For all other years, I calculate the percent with no access to the internet as the count of the population reporting no internet access divided by the total population. Thus, I am able to retrieve the effect of compact participation within some

<sup>3</sup>Effective dates for IMLC and PSYPACT adoption are listed in Appendix Table A1.

level of access to internet. These state-level variables are included in my main specification for all outcomes.

In addition to demographic controls from ACS, I include an indicator for state telehealth payment parity legislation. Payment parity is associated with higher levels of telehealth usage (Erikson et al., 2022; Harvey et al., 2019; Lee and Singh, 2023; Neufeld et al., 2016). Therefore, I gather and compile the effective dates of payment parity requirements for private insurers from each state’s respective statute(s) to control for this relationship.

Data used for individual controls such as health insurance status and demographics are included in the models with BRFSS outcomes. I include demographic controls for gender, race, age, education level, income, employment status, and marital status in addition to a control for having health insurance. Because the highest income tier recorded by BRFSS at the start of my analysis period includes all individuals who report having an income greater than \$75,000, all individuals falling in higher income tiers in the following survey years are placed in this tier. These variables are only applicable and included in the analysis of poor mental health days and the incidence of MDD.

**Table 1:** Summary Statistics

|                                                                | BRFSS – All<br>Respondents | BRFSS – Ever<br>Diagnosed with<br>Depression | State-level       |
|----------------------------------------------------------------|----------------------------|----------------------------------------------|-------------------|
| <i>Mental Health Outcomes</i>                                  |                            |                                              |                   |
| Poor mental health days                                        | 3.701<br>(7.935)           | 10.583<br>(11.405)                           |                   |
| Major depressive disorder ( $\geq 14$ poor mental health days) | 0.115<br>(0.319)           | 0.360<br>(0.480)                             |                   |
| Major depressive disorder ( $\geq 21$ poor mental health days) | 0.066<br>(0.248)           | 0.220<br>(0.414)                             |                   |
| Age-adjusted suicide rate per 100,000                          |                            |                                              | 15.621<br>(4.754) |
| SSRI prescribing rate, Medicare population                     |                            |                                              | 0.031<br>(0.005)  |
| <i>Compact Participation</i>                                   |                            |                                              |                   |
| IMLC active                                                    | 0.274<br>(0.446)           | 0.287<br>(0.452)                             | 0.281*<br>(0.450) |
| PSYPACT active                                                 | 0.133<br>(0.339)           | 0.143<br>(0.350)                             | 0.127*<br>(0.333) |
| Either compact active                                          | 0.304<br>(0.460)           | 0.318<br>(0.466)                             | 0.311*<br>(0.463) |
| Observations                                                   | 5,684,984                  | 1,077,380                                    | 663*              |

*Note:* Means are shown with standard deviations in parentheses. \* State-level compact participation statistics and the reported number of observations correspond to the suicide-rate sample ( $N = 663$ ). The SSRI prescribing-rate sample contains 510 state-year observations.

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## 4 Methods

The analysis is split into three parts. First, I examine whether interstate licensure compact participation increases multi-state licensure among eligible providers. Second, I estimate the effects of compact participation on individual-level mental health outcomes. Finally, I examine effects on state-level mental health outcomes.

I begin by estimating the effect of compact participation on provider licensure using the provider-year panel described above. The baseline specification is:

$$LicensedStates_{ist} = \beta_0 + \beta_1 Compact_{st} + \gamma_i + \delta_t + \epsilon_{ist} \quad (1)$$

where  $LicensedStates_{ist}$  is the number of states in which provider  $i$  with origin state  $s$  holds an active license in year  $t$ .  $Compact_{st}$  is an indicator equal to one when the provider's origin state participates in the compact for which the provider is eligible. I use IMLC participation for primary care physicians and psychiatrists and PSYPACT participation for psychologists. Provider and year fixed effects are represented by  $\gamma_i$  and  $\delta_t$ , respectively. Standard errors are clustered at the provider level.

I estimate Equation 1 using ordinary least squares. I then replace  $LicensedStates_{ist}$  with  $Multistate_{ist}$ , an indicator equal to one if provider  $i$  holds active licenses in more than one state in year  $t$ , and estimate a linear probability model. The provider fixed effects account for time-invariant differences across providers, while the year fixed effects account for common changes in licensure over time.

I supplement these estimates with ordered logit models that categorize providers as licensed in one, two, three, or four or more states and report average marginal effects for each category. I also descriptively examine annual transitions across licensure categories, comparing transition rates based on whether the provider's origin state participates in the relevant compact.

Next, I examine the effects of interstate licensure compacts on individual-level mental health outcomes. The equation used to estimate the effects of compact participation on the count of poor mental health days in the last month is given by:

$$PMHD_{ist} = \exp(\beta_0 + \beta_1 IMLC_{st} + \beta_2 PSYPACT_{st} + \beta_3 X_{st} + \beta_4 Q_{ist} + \gamma_s + \delta_t + \varepsilon_{ist}) \quad (2)$$

where  $PMHD_{ist}$  represents the count of days of poor mental health in the last 30 days for individual  $i$  in state  $s$  at time  $t$ .  $IMLC_{st}$  and  $PSYPACT_{st}$  are indicators for participation in IMLC and/or PSYPACT in state  $s$  at time  $t$ , respectively. The vector  $X_{st}$  consists of demographic controls at the state level as well as the percent of the population without access to internet and an indicator for telehealth payment parity legislation, and  $Q_{ist}$  consists of individual-level demographic controls. State and year fixed effects are captured by  $\gamma_s$  and  $\delta_t$ , respectively.

The percent change in the count of poor mental health days associated with participation in the IMLC and PSYPACT is given by  $\exp(\beta_1) - 1$  and  $\exp(\beta_2) - 1$ , respectively. I estimate this effect using a Poisson regression with a ratio-in-ratios framework with standard errors clustered by state.

The equation used to estimate the effects of compact participation on the incidence of major depressive disorder is given by:

$$MDD_{ist} = \beta_0 + \beta_1 IMLC_{st} + \beta_2 PSYPACT_{st} + \beta_3 X_{st} + \beta_4 Q_{ist} + \gamma_s + \delta_t + \varepsilon_{ist} \quad (3)$$

where  $MDD_{ist}$  is an indicator for individual  $i$  having major depressive disorder, as defined by either 14 or more poor mental health days in the last month or 21 or more poor mental health days in the last month, in state  $s$  at time  $t$ . All other variables are defined as before.

I estimate Equation 3 with linear probability models with standard errors clustered by state. Thus, the coefficients of interest,  $\beta_1$  and  $\beta_2$ , represent the change in the probability of individual  $i$  experiencing major depressive disorder after state  $s$  participates in the IMLC or PSYPACT.

Finally, I examine the effects of interstate licensure compacts on the state-level mental health outcomes described above. The equation used to estimate these effects is given by:

$$MH_{st} = \beta_0 + \beta_1 IMLC_{st} + \beta_2 PSYPACT_{st} + \beta_3 X_{st} + \gamma_s + \delta_t + \varepsilon_{st} \quad (4)$$

where  $MH_{st}$  represents the suicide rate or the Medicare prescription rate of the top five SSRIs in state  $s$  at time  $t$ . Again,  $IMLC_{st}$  and  $PSYPACT_{st}$  are indicators for participation in the IMLC and/or PSYPACT in state  $s$  at time  $t$ . The vector  $X_{st}$  consists of demographic controls as well as the percent of the population without access to internet and an indicator for telehealth payment parity, and  $\gamma_s$  and  $\delta_t$  capture state and year fixed effects. The coefficients on  $IMLC_{st}$  and  $PSYPACT_{st}$  represent the change in the suicide or SSRI prescribing rate after state  $s$  begins participating in the respective interstate compact.

In addition to the specifications outlined above, I include models with a single treatment variable defined as state participation in at least one of these interstate compacts. The estimates from these models represent the effects of streamlined licensure regardless of which mental health professionals are directly impacted.

The validity of the difference-in-differences coefficients relies on the presence of parallel trends. In order to address a potential violation of this assumption, I conduct event studies in which the binary indicators for compact participation are replaced with relative time indicators for the periods leading to and following participation. This allows me to check for differing trends in mental health between participating and non-participating states. These event studies also show if effects occur with a lag.

An additional threat to the validity of the difference-in-differences model estimates is the staggered timing of policy adoption and heterogeneous treatment effects. To address this potential issue, I estimate additional event studies using the framework by Sun and Abraham (2021). These authors suggest a decomposition of event study methods in which relative time indicators are saturated with indicators for treatment year group and aggregated to relative time indicators by cohort size. This method keeps earlier treated observations from contaminating the control group.

It is possible that states enter interstate licensure compacts in response to its populations poor mental health. This reverse causality is not an issue for my individual-level analysis. The state-level analyses are, however, vulnerable to this threat. Because the IMLC is applicable for all medical doctors and doctors of osteopathic medicine, not just mental health professionals, it is unlikely that a state passes legislation to enter this compact in response to its population's mental health outcomes. On the other hand, because PSYPACT is only applicable for psychologists, a state's participation in this compact may be a response to its population's poor mental health outcomes. This is not applicable for the Medicare SSRI prescription rate as psychologists cannot prescribe medications. However, it is possible that a state's suicide rate may impact the decision to enter PSYPACT. I use the pre-treatment event study estimates to assess whether trends differ between treated and control states prior to compact participation.

Because adoption of compact legislation coincides with the COVID-19 pandemic, I show that the effects of IMLC participation over a shortened period from 2011 to 2019 agree with those from the entire study period. Since PSYPACT was not active until mid-2020, this same robustness check cannot be performed. To further ensure the validity of my findings, I also test alternative specifications that include controls for temporary expansions in telehealth and out-of-state license

recognition. As shown in Appendix Tables A6–A8, the results remain robust even after accounting for these temporary healthcare policy changes.

## 5 Results

### 5.1 Effects of Compacts on Multi-state Licensure

Increasing multi-state licensure is the direct intended effect of these policies and is a prerequisite for compacts to expand access to care across state lines. I, therefore, first examine whether providers respond to streamlined licensing by obtaining licenses in additional states.

Table 2 presents estimates of the effect of interstate licensure compact participation on provider licensure. Panel A examines the number of states in which a provider holds an active license, while Panel B examines the probability that a provider holds licenses in more than one state. IMLC participation increases the number of licensed states by 0.0051 among primary care physicians and 0.0087 among psychiatrists. PSYPACT participation is associated with an increase of 0.0019 states among psychologists, although the estimate is only marginally significant.

The linear probability estimates similarly indicate that compact participation increases the probability of multi-state licensure by 0.45 percentage points among primary care physicians and 0.56 percentage points among psychiatrists, relative to baseline rates of 3.9 and 4.0 percent, respectively. PSYPACT participation increases the probability of multi-state licensure among psychologists by 0.18 percentage points relative to a baseline rate of 1.4 percent. These estimates correspond to increases of approximately 12 to 14 percent relative to the respective baseline rates, indicating meaningful changes in multi-state licensure despite relatively small percentage-point effects.

Table 3 provides complementary descriptive evidence on annual changes in the number of licenses held. The most common upward transition is from one to two licensed states. Among physicians beginning a year with one state license, 1.40 percent transition to two licenses when the IMLC is active, compared with 1.06 percent when it is not active. Among psychologists beginning a year with one license, 0.45 percent transition to two licenses when PSYPACT is active, compared with 0.35 percent when it is not active. Transitions directly from one state to three or four or more states are uncommon under both compacts. Among providers already licensed in multiple states, upward transitions also occur, although these transitions are considerably less common in absolute terms and differences by compact participation are less consistent. These descriptive patterns indicate that differences in licensure transitions associated with compact participation are concentrated primarily among providers adding a second state license. Ordered logit estimates reported in Appendix Table A9 corroborate this pattern, with compact participation primarily shifting the probability of licensure from the one-state to the two-state category and substantially smaller changes in the probability of holding three or four or more licenses.

These findings provide evidence that interstate licensure compacts affect provider behavior in the direction intended by the policies. Eligible providers respond to streamlined licensing by obtaining licenses in additional states, creating the potential for expanded access to care across state lines. The remainder of the analysis examines whether this supply-side response translates into measurable changes in population mental health outcomes.

### 5.2 Effects of Compacts on Poor Mental Health Days

Table 4 shows estimation results of the effects of interstate compact participation on the count of monthly poor mental health days. Column 1 presents estimates for the main specification as

**Table 2:** Effects of Interstate Licensure Compacts on Provider Licensure

|                                           | Primary Care<br>(1)   | Psychiatry<br>(2) | Psychology<br>(3)    |
|-------------------------------------------|-----------------------|-------------------|----------------------|
| <i>Panel A: Number of Licensed States</i> |                       |                   | [2pt] IMLC           |
| 0.0051***                                 | 0.0087***<br>(0.0008) | (0.0032)          |                      |
| PSYPACT                                   |                       |                   | 0.0019*<br>(0.0010)  |
| Observations                              | 1,508,581             | 151,355           | 168,923              |
| Mean outcome                              | 1.044                 | 1.047             | 1.015                |
| <i>Panel B: Multi-state Licensure</i>     |                       |                   | [2pt] IMLC           |
| 0.0045***                                 | 0.0056***<br>(0.0005) | (0.0018)          |                      |
| PSYPACT                                   |                       |                   | 0.0018**<br>(0.0009) |
| Observations                              | 1,508,581             | 151,355           | 168,923              |
| Mean outcome                              | 0.039                 | 0.040             | 0.014                |

*Notes:* Panel A examines the number of states in which a provider holds an active license. Panel B examines an indicator equal to one if a provider holds licenses in more than one state. Columns (1) and (2) estimate the effect of IMLC participation among primary care physicians and psychiatrists, respectively. Column (3) estimates the effect of PSYPACT participation among psychologists. All specifications include provider and year fixed effects. Standard errors are clustered at the provider level. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.10$ .

**Table 3:** Transitions in Number of Licensed States

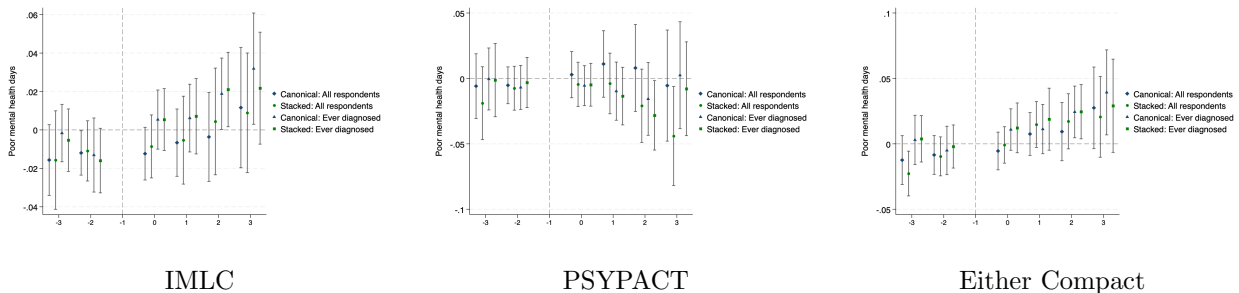
| Transition                                                       | Compact Not Active |         | Compact Active |         |
|------------------------------------------------------------------|--------------------|---------|----------------|---------|
|                                                                  | Number             | Percent | Number         | Percent |
| <i>Panel A: IMLC — Primary Care Physicians and Psychiatrists</i> |                    |         |                |         |
| 1 → 2                                                            | 8,941              | 1.06    | 6,166          | 1.40    |
| 1 → 3                                                            | 216                | 0.03    | 162            | 0.04    |
| 1 → 4+                                                           | 79                 | 0.01    | 44             | 0.01    |
| 2 → 3                                                            | 455                | 2.13    | 460            | 2.59    |
| 2 → 4+                                                           | 39                 | 0.18    | 28             | 0.16    |
| 3 → 4+                                                           | 38                 | 3.33    | 47             | 4.32    |
| <i>Panel B: PSYPACT — Psychologists</i>                          |                    |         |                |         |
| 1 → 2                                                            | 368                | 0.35    | 125            | 0.45    |
| 1 → 3                                                            | 11                 | 0.01    | 4              | 0.01    |
| 1 → 4+                                                           | 3                  | 0.00    | 0              | 0.00    |
| 2 → 3                                                            | 11                 | 1.24    | 8              | 1.43    |
| 2 → 4+                                                           | 0                  | 0.00    | 0              | 0.00    |
| 3 → 4+                                                           | 0                  | 0.00    | 1              | 3.70    |

*Notes:* The table reports annual transitions in the number of states in which a provider holds an active license. Panel A reports transitions among primary care physicians and psychiatrists according to whether the provider's origin state participates in the IMLC. Panel B reports transitions among psychologists according to whether the provider's origin state participates in PSYPACT. The number columns report the number of observed transitions. The percent columns report the percentage of provider-years beginning in the indicated license category that transition to the destination category in the following year. Licensure counts of four or more states are grouped into a single category. Providers without an observed licensure count in the preceding year are excluded.

described in Equation 2 over the entire study period. Column 2 shows the impact of participation in either compact on poor mental health days. Columns 3 and 4 represent estimates for the effects of IMLC participation over the periods 2011 to 2019 and 2011 to 2023, respectively. Panel A presents estimates for my entire sample population while Panel B shows those for the population ever diagnosed with depression only.

I find that participation in the IMLC has no effect on poor mental health days on average for the entire sample, but participation increases this count for those diagnosed with depression by 1.8 percent, or 0.19 days. The coefficients for PSYPACT show that participation does not lead to reductions in self-reported poor mental health days. In fact, PSYPACT participation increases reported poor mental health days by around 0.1 days on a base of 3.70 days. The number of poor mental health days experienced each month on average increases by about 2 percent following participation in either the IMLC or PSYPACT. While the direction of these point estimates is positive, the estimated increases are small in magnitude. Estimates for Columns 3 and 4 show that results are robust to including the years 2020 to 2023. Thus, results seem to be unaffected by the COVID-19 pandemic. As an additional robustness check, I estimate the effects of compact participation on poor mental health days using the Wooldridge (2021) DID estimator with a Poisson framework. These results are presented in Appendix Table A3 and similarly provide no evidence that compact participation reduces poor mental health days.

Figure 2 presents event study plots detailing average poor mental health days both before and after different compact legislation implementation. These estimates echo those in Table 4, showing little-to-no impact of compact participation on poor mental health days. While this lack of effect is persistent over time, point estimates are increasing, consistent with the average treatment effects.



**Figure 2:** Event Studies for Poor Mental Health Days across Compacts

### 5.3 Effects of Compacts on Major Depressive Disorder

Estimation results for the effects of interstate compact participation on the incidence of major depressive disorder are presented in Table 5. Again, Panel A presents estimates for my entire sample population while Panel B shows those for the population ever diagnosed with depression only. Columns 1-4 show the impacts on MDD defined as 14 or more poor mental health days in the last month (MDD14), while Columns 5-8 show those for MDD defined as 21 or more poor mental health days in the last month (MDD21). Columns 3 and 7 report coefficients for the period 2011 to 2019. All others provide estimates over the entire study period.

The probability of experiencing MDD14 increases by between 0.2 and 0.7 percentage points following participation in either the IMLC or PSYPACT. Again, however, the small magnitude of these increases should be considered. I find no effects of participation in either interstate compact

**Table 4:** Poor Mental Health Days and Compact Participation

|                                                | 2011–2023                     |                               | 2011–2019                    | 2011–2023                     |
|------------------------------------------------|-------------------------------|-------------------------------|------------------------------|-------------------------------|
|                                                | (1)                           | (2)                           | (3)                          | (4)                           |
| <i>Panel A: All Respondents</i>                |                               |                               |                              |                               |
| IMLC                                           | 0.012<br>(0.010)<br>[0.045]   |                               | 0.010<br>(0.009)<br>[0.034]  | 0.017<br>(0.010)<br>[0.062]   |
| PSYPACT                                        | 0.027**<br>(0.011)<br>[0.098] |                               |                              |                               |
| Either compact                                 |                               | 0.021**<br>(0.009)<br>[0.076] |                              |                               |
| Percent no access to internet                  | 0.453**<br>(0.186)            | 0.455**<br>(0.192)            | 0.070<br>(0.145)             | 0.438**<br>(0.194)            |
| Telehealth payment parity                      | -0.007<br>(0.009)             | -0.007<br>(0.009)             | -0.010<br>(0.011)            | -0.007<br>(0.009)             |
| Observations                                   | 5,683,627                     | 5,683,627                     | 4,028,804                    | 5,683,627                     |
| Mean dependent variable                        | 3.70                          | 3.70                          | 3.49                         | 3.70                          |
| <i>Panel B: Ever Diagnosed with Depression</i> |                               |                               |                              |                               |
| IMLC                                           | 0.018**<br>(0.009)<br>[0.190] |                               | 0.014*<br>(0.007)<br>[0.141] | 0.018**<br>(0.009)<br>[0.194] |
| PSYPACT                                        | 0.003<br>(0.010)<br>[0.029]   |                               |                              |                               |
| Either compact                                 |                               | 0.019**<br>(0.008)<br>[0.196] |                              |                               |
| Percent no access to internet                  | 0.274**<br>(0.134)            | 0.289**<br>(0.131)            | 0.073<br>(0.138)             | 0.273**<br>(0.135)            |
| Telehealth payment parity                      | -0.002<br>(0.009)             | -0.001<br>(0.009)             | 0.002<br>(0.009)             | -0.002<br>(0.009)             |
| Observations                                   | 1,077,112                     | 1,077,112                     | 749,282                      | 1,077,112                     |
| Mean dependent variable                        | 10.58                         | 10.58                         | 10.33                        | 10.58                         |

*Notes:* Estimates in brackets show marginal effects. Standard errors are clustered by state. All models include state and year fixed effects and all control variables outlined in the Data section. Full model coefficient estimates are presented in Appendix Table A2. \*\*\* $p < 0.01$ , \*\* $p < 0.05$ , \* $p < 0.10$ .

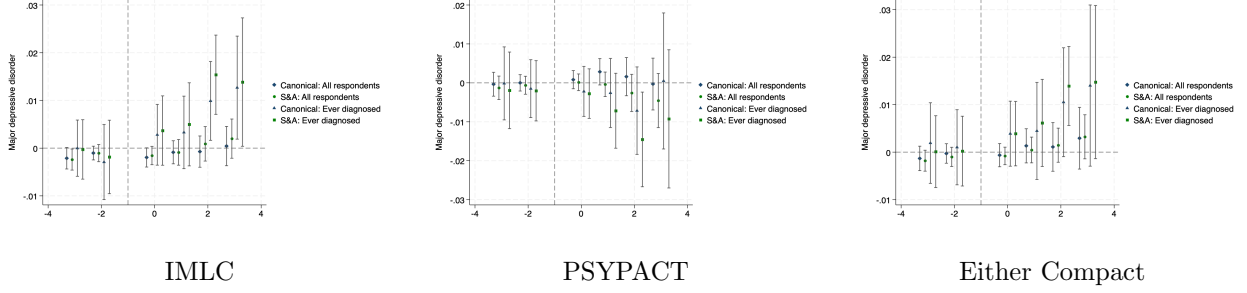
**Table 5:** Major Depressive Disorder and Compact Participation

|                                                | Major Depressive Disorder ( $\geq 14$ days) |                    |                   |                    | Major Depressive Disorder ( $\geq 21$ days) |                    |                   |                    |
|------------------------------------------------|---------------------------------------------|--------------------|-------------------|--------------------|---------------------------------------------|--------------------|-------------------|--------------------|
|                                                | 2011–2023<br>(1)                            | 2011–2023<br>(2)   | 2011–2019<br>(3)  | 2011–2023<br>(4)   | 2011–2023<br>(5)                            | 2011–2023<br>(6)   | 2011–2019<br>(7)  | 2011–2023<br>(8)   |
| <i>Panel A: All Respondents</i>                |                                             |                    |                   |                    |                                             |                    |                   |                    |
| IMLC                                           | 0.001<br>(0.001)                            |                    | 0.001<br>(0.001)  | 0.002<br>(0.001)   | 0.000<br>(0.001)                            |                    | 0.000<br>(0.001)  | 0.000<br>(0.001)   |
| PSYPACT                                        | 0.004**<br>(0.002)                          |                    |                   |                    | 0.002**<br>(0.001)                          |                    |                   |                    |
| Either compact                                 |                                             | 0.002*<br>(0.001)  |                   |                    |                                             | 0.000<br>(0.001)   |                   |                    |
| Percent no access to internet                  | 0.019<br>(0.024)                            | 0.018<br>(0.025)   | -0.012<br>(0.016) | 0.017<br>(0.025)   | 0.002<br>(0.012)                            | 0.002<br>(0.013)   | -0.016<br>(0.010) | 0.001<br>(0.013)   |
| Telehealth payment parity                      | -0.001<br>(0.001)                           | -0.001<br>(0.001)  | -0.002<br>(0.001) | -0.001<br>(0.001)  | -0.001*<br>(0.001)                          | -0.001*<br>(0.001) | -0.001<br>(0.001) | -0.002*<br>(0.001) |
| Observations                                   | 5,683,627                                   | 5,683,627          | 4,028,804         | 5,683,627          | 5,683,627                                   | 5,683,627          | 4,028,804         | 5,683,627          |
| R <sup>2</sup>                                 | 0.081                                       | 0.081              | 0.084             | 0.081              | 0.058                                       | 0.058              | 0.063             | 0.058              |
| Mean dependent variable                        | 0.115                                       | 0.115              | 0.109             | 0.115              | 0.066                                       | 0.066              | 0.064             | 0.066              |
| <i>Panel B: Ever Diagnosed with Depression</i> |                                             |                    |                   |                    |                                             |                    |                   |                    |
| IMLC                                           | 0.007**<br>(0.003)                          |                    | 0.006*<br>(0.003) | 0.007**<br>(0.003) | 0.002<br>(0.002)                            |                    | 0.000<br>(0.002)  | 0.002<br>(0.002)   |
| PSYPACT                                        | 0.001<br>(0.004)                            |                    |                   |                    | 0.001<br>(0.003)                            |                    |                   |                    |
| Either compact                                 |                                             | 0.007**<br>(0.003) |                   |                    |                                             | 0.002<br>(0.002)   |                   |                    |
| Percent no access to internet                  | 0.066<br>(0.049)                            | 0.072<br>(0.048)   | -0.001<br>(0.057) | 0.066<br>(0.049)   | 0.031<br>(0.035)                            | 0.032<br>(0.035)   | -0.008<br>(0.042) | 0.030<br>(0.036)   |
| Telehealth payment parity                      | -0.001<br>(0.003)                           | -0.001<br>(0.004)  | -0.001<br>(0.004) | -0.001<br>(0.003)  | -0.003<br>(0.002)                           | -0.003<br>(0.002)  | 0.000<br>(0.003)  | -0.003<br>(0.002)  |
| Observations                                   | 1,077,112                                   | 1,077,112          | 749,282           | 1,077,112          | 1,077,112                                   | 1,077,112          | 749,282           | 1,077,112          |
| R <sup>2</sup>                                 | 0.097                                       | 0.097              | 0.105             | 0.097              | 0.075                                       | 0.075              | 0.082             | 0.075              |
| Mean dependent variable                        | 0.360                                       | 0.360              | 0.350             | 0.360              | 0.220                                       | 0.220              | 0.218             | 0.220              |

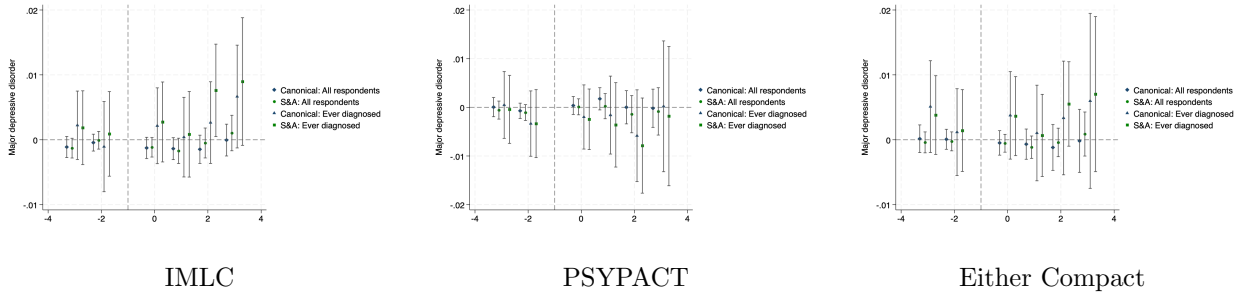
*Notes:* Standard errors are clustered by state. All models include state and year fixed effects and all control variables outlined in the Data section. Full model coefficient estimates are presented in Appendix Table A4. \*\*\* $p < 0.01$ , \*\* $p < 0.05$ , \* $p < 0.10$ .

on the incidence of MDD21. Since the true incidence of MDD in the sample population is likely to fall between the estimates of MDD14 and MDD21, the estimates in Table 5 imply that compact participation has no substantial effect on the incidence of major depressive disorder.

Figures 3 and 4 present event study plots for MDD14 and MDD21, respectively. These estimates show no impact of compact participation on the incidence of major depressive disorder, raising caution in interpreting the significant point estimates in the main specification.



**Figure 3:** Event Studies for MDD14 across Compacts



**Figure 4:** Event Studies for MDD21 across Compacts

## 5.4 Effects of Compacts on State-Level Mental Health

Table 6 shows the estimation results from Equation 4. Columns 1-4 show the impacts on age adjusted suicide rate. Columns 5 and 6 show those for the Medicare SSRI prescribing rate. Columns 3 and 5 report coefficients for the period before 2020.

I find that PSYPACT participation marginally reduces the suicide rate in participating states. This effect, however, represents only a 0.4 reduction on a mean of 15.6 suicides per 100,000 people. Estimates show that participation in the IMLC, on the other hand, is associated with an increase in suicide rate of a similar magnitude.

Because PSYPACT streamlines licensure exclusively for psychologists, who cannot prescribe medication, I report only estimates of the effects of IMLC participation on the prescribing of SSRIs in Table 6. Columns 5 and 6 show that participation in the IMLC has no effect on the SSRI prescribing rate. I also run PSYPACT specifications on this outcome as a falsification test. These estimates confirm that there is no effect of PSYPACT participation on the Medicare SSRI prescribing rate.

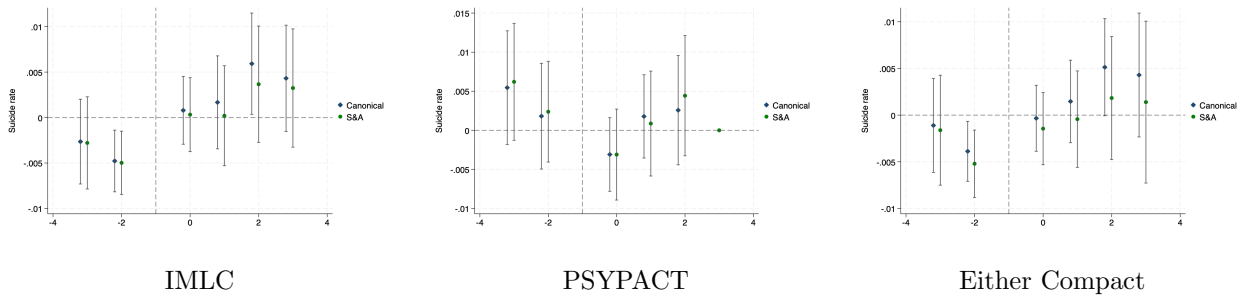
Figures 5 and 6 present event study plots for state suicide rate and Medicare SSRI prescribing rate, respectively. Although the main models suggest statistically significant effects on suicide

**Table 6:** State-level Mental Health and Compact Participation

|                               | Suicide Rate per 100,000 |                   |                    |                    | SSRI Prescribing Rate |                   |
|-------------------------------|--------------------------|-------------------|--------------------|--------------------|-----------------------|-------------------|
|                               | 2011–2023<br>(1)         | 2011–2019<br>(2)  | 2011–2023<br>(3)   | 2011–2023<br>(4)   | 2013–2019<br>(5)      | 2013–2022<br>(6)  |
| IMLC                          | 0.502**<br>(0.215)       |                   | 0.155<br>(0.198)   | 0.449**<br>(0.216) | 0.000<br>(0.000)      | 0.000<br>(0.000)  |
| PSYPACT                       | -0.388*<br>(0.228)       |                   |                    |                    |                       |                   |
| Either compact                |                          | 0.329<br>(0.210)  |                    |                    |                       |                   |
| Percent no access to internet | -0.511<br>(4.423)        | -0.354<br>(4.391) | 7.400**<br>(3.400) | -0.460<br>(4.487)  | -0.011<br>(0.009)     | -0.010<br>(0.012) |
| Telehealth payment parity     | -0.270<br>(0.256)        | -0.283<br>(0.265) | -0.397<br>(0.363)  | -0.277<br>(0.262)  | 0.000<br>(0.000)      | 0.000<br>(0.000)  |
| Observations                  | 663                      | 663               | 459                | 663                | 357                   | 510               |
| R <sup>2</sup>                | 0.954                    | 0.954             | 0.958              | 0.954              | 0.993                 | 0.987             |
| Mean dependent variable       | 15.62                    | 15.62             | 15.28              | 15.62              | 0.031                 | 0.031             |

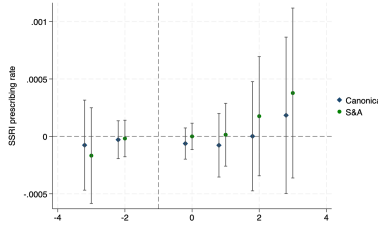
*Notes:* Standard errors are clustered by state. All models include state and year fixed effects and all control variables outlined in the Data section. Full model coefficient estimates are presented in Appendix Table A5. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.10$ .

rates, these estimates provide little evidence of persistent post-participation changes in either state-level outcome. The event study results in Figure 5 also indicate a violation of the parallel trends assumption, potentially biasing the point estimates reported above.

**Figure 5:** Event Studies for Suicide Rate across Compacts

## 6 Conclusion

Interstate licensure compacts reduce administrative barriers to practicing across state lines with the goal of expanding access to care. I find that providers respond to these policies by obtaining licenses in additional states. IMLC participation increases the probability of multi-state licensure among primary care physicians and psychiatrists, while PSYPACT participation increases multi-state licensure among psychologists. These effects are meaningful relative to the low baseline rate of



**Figure 6:** Event Studies for SSRI Prescribing Rate across Compacts

multi-state licensure, providing evidence that the compacts affect provider behavior in the direction intended by the policies.

Despite this increase in multi-state licensure, I find no consistent evidence that compact participation improves population mental health outcomes. Compact participation does not reduce the number of poor mental health days or the incidence of major depressive disorder. Some estimates instead indicate small increases in poor mental health days. At the state level, IMLC participation has no statistically significant effect on SSRI prescribing rates among Medicare beneficiaries. Estimates indicate that PSYPACT participation is associated with a marginal reduction in suicide rates, while IMLC participation corresponds with an increase of similar magnitude. However, event study results suggest a violation of the parallel trends assumption for suicide rates, raising caution when interpreting these estimates.

These findings suggest that reducing administrative barriers to interstate practice can increase multi-state licensure, but that expanded licensure alone may be insufficient to improve population mental health outcomes. Obtaining a license in an additional state creates the opportunity for a provider to serve patients across state lines, but does not necessarily imply that the provider increases the total amount of care delivered or that increased multi-state practice translates into improvements in patient mental health. From a policy perspective, interstate licensure compacts may, therefore, need to be paired with additional interventions that address other barriers to accessing mental health care.

Why increased multi-state licensure does not translate into measurable improvements in population mental health remains an important question. Providers may reallocate their existing time across a broader geographic patient base, alter the amount of time spent with each patient, or shift toward different modes of care, such as telehealth, without necessarily increasing the total amount of care they provide. Alternatively, barriers beyond state licensure may limit the extent to which the opportunity to practice across state lines translates into greater access to care. The data used in this analysis do not allow me to distinguish among these potential mechanisms.

Future research should focus on understanding how compact participation affects provider-level decisions, such as changes in patient caseloads, treatment intensity, or geographic distribution of care. Detailed data on visit volume, visit length, treatment quality, and patient characteristics before and after compact implementation would help clarify how increased multi-state licensure translates into changes in care delivery. In particular, evaluating how these effects vary by provider type, specialty, or rural versus urban areas could reveal important heterogeneity and help explain why increased multi-state licensure does not necessarily translate into improvements in population health.

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## Appendix

**Table A1:** Active Compact Dates

| State    | IMLC   | PSYPACT  |
|----------|--------|----------|
| Alabama  | 4/1/17 | 6/1/21   |
| Alaska   |        |          |
| Arizona  | 4/1/17 | 7/1/20   |
| Arkansas |        | 11/18/21 |

|                      |          |          |
|----------------------|----------|----------|
| California           |          |          |
| Colorado             | 4/1/17   | 7/1/20   |
| Connecticut          | 5/13/22  | 10/1/22  |
| Delaware             | 6/23/21  | 7/1/20   |
| District of Columbia | 6/12/23  | 4/2/21   |
| Florida              |          | 7/1/23   |
| Georgia              | 9/16/21  | 7/1/20   |
| Hawaii               |          |          |
| Idaho                | 4/1/17   | 7/1/22   |
| Illinois             | 4/1/17   | 7/1/20   |
| Indiana              | 5/1/23   | 7/1/22   |
| Iowa                 | 4/1/17   |          |
| Kansas               | 4/1/17   | 1/1/22   |
| Kentucky             | 12/13/18 | 6/28/21  |
| Louisiana            | 7/1/21   |          |
| Maine                | 4/6/17   | 10/18/21 |
| Maryland             | 1/19/18  | 5/18/21  |
| Massachusetts        |          |          |
| Michigan             | 6/1/19   | 3/29/23  |
| Minnesota            | 4/1/17   | 5/26/21  |
| Mississippi          | 4/1/17   |          |
| Missouri             |          | 7/1/20   |
| Montana              | 4/1/17   |          |
| Nebraska             | 4/1/17   | 7/1/20   |
| Nevada               | 4/1/17   | 7/1/20   |
| New Hampshire        | 4/1/17   | 7/1/20   |
| New Jersey           |          | 11/23/21 |
| New Mexico           |          |          |
| New York             |          |          |
| North Carolina       |          | 3/1/21   |
| North Dakota         | 1/15/19  | 8/1/23   |
| Ohio                 | 7/1/21   | 7/26/21  |
| Oklahoma             | 1/18/19  | 7/1/20   |
| Oregon               |          |          |
| Pennsylvania         |          | 7/8/20   |
| Rhode Island         |          | 7/1/23   |
| South Carolina       |          | 7/17/23  |
| South Dakota         | 4/1/17   |          |
| Tennessee            | 4/1/17   | 5/11/21  |
| Texas                | 6/7/22   | 7/1/20   |
| Utah                 | 4/1/17   | 7/1/20   |
| Vermont              | 12/21/17 |          |
| Virginia             |          | 1/1/21   |
| Washington           | 4/1/17   | 6/9/22   |
| West Virginia        | 4/1/17   | 11/18/21 |
| Wisconsin            | 4/1/17   | 2/6/22   |

**Table A2:** Poor Mental Health Days and Compact Participation

|                                 | 2011–2023                     |                               | 2011–2019                   | 2011–2023                   |
|---------------------------------|-------------------------------|-------------------------------|-----------------------------|-----------------------------|
|                                 | (1)                           | (2)                           | (3)                         | (4)                         |
| <i>Panel A: All Respondents</i> |                               |                               |                             |                             |
| IMLC                            | 0.012<br>(0.010)<br>[0.045]   |                               | 0.010<br>(0.009)<br>[0.034] | 0.017<br>(0.010)<br>[0.062] |
| PSYPACT                         | 0.027**<br>(0.011)<br>[0.098] |                               |                             |                             |
| Either compact                  |                               | 0.021**<br>(0.009)<br>[0.076] |                             |                             |
| Male                            | -0.289***<br>(0.007)          | -0.289***<br>(0.007)          | -0.269***<br>(0.007)        | -0.289***<br>(0.007)        |
| Race and ethnicity              |                               |                               |                             |                             |
| Black                           | -0.236***<br>(0.013)          | -0.236***<br>(0.013)          | -0.246***<br>(0.015)        | -0.236***<br>(0.013)        |
| Other                           | -0.026<br>(0.018)             | -0.026<br>(0.018)             | -0.005<br>(0.018)           | -0.026<br>(0.018)           |
| Hispanic                        | -0.240***<br>(0.018)          | -0.240***<br>(0.018)          | -0.217***<br>(0.019)        | -0.240***<br>(0.018)        |
| Uninsured                       | -0.003<br>(0.011)             | -0.003<br>(0.011)             | 0.007<br>(0.010)            | -0.003<br>(0.011)           |
| Married                         | -0.196***<br>(0.012)          | -0.196***<br>(0.012)          | -0.157***<br>(0.012)        | -0.196***<br>(0.012)        |
| Divorced                        | 0.131***<br>(0.006)           | 0.131***<br>(0.006)           | 0.149***<br>(0.007)         | 0.131***<br>(0.006)         |
| Widowed                         | -0.005<br>(0.009)             | -0.005<br>(0.009)             | 0.008<br>(0.010)            | -0.005<br>(0.009)           |
| Age                             |                               |                               |                             |                             |
| 35–44                           | -0.064***<br>(0.006)          | -0.064***<br>(0.006)          | -0.026***<br>(0.006)        | -0.064***<br>(0.006)        |
| 45–54                           | -0.163***<br>(0.010)          | -0.163***<br>(0.010)          | -0.098***<br>(0.009)        | -0.163***<br>(0.010)        |
| 55–64                           | -0.353***<br>(0.013)          | -0.353***<br>(0.013)          | -0.273***<br>(0.013)        | -0.353***<br>(0.013)        |
| 65+                             | -0.741***<br>(0.016)          | -0.741***<br>(0.016)          | -0.679***<br>(0.016)        | -0.741***<br>(0.016)        |
| High school or less             | 0.082***                      | 0.082***                      | 0.081***                    | 0.082***                    |

|                                 | 2011–2023 |           | 2011–2019 | 2011–2023 |
|---------------------------------|-----------|-----------|-----------|-----------|
|                                 | (1)       | (2)       | (3)       | (4)       |
|                                 | (0.017)   | (0.017)   | (0.020)   | (0.017)   |
| Some college/Associate's degree | 0.117***  | 0.116***  | 0.105***  | 0.116***  |
|                                 | (0.017)   | (0.017)   | (0.021)   | (0.017)   |
| Bachelor's degree or higher     | -0.067*** | -0.067*** | -0.086*** | -0.067*** |
|                                 | (0.019)   | (0.019)   | (0.021)   | (0.019)   |
| Income                          |           |           |           |           |
| \$15,000–25,000                 | 0.580***  | 0.580***  | 0.638***  | 0.580***  |
|                                 | (0.014)   | (0.014)   | (0.015)   | (0.014)   |
| \$25,000–35,000                 | 0.472***  | 0.472***  | 0.530***  | 0.472***  |
|                                 | (0.014)   | (0.014)   | (0.015)   | (0.014)   |
| \$35,000–50,000                 | 0.357***  | 0.357***  | 0.404***  | 0.357***  |
|                                 | (0.012)   | (0.012)   | (0.012)   | (0.012)   |
| \$50,000–75,000                 | 0.277***  | 0.277***  | 0.313***  | 0.277***  |
|                                 | (0.009)   | (0.009)   | (0.009)   | (0.009)   |
| \$75,000+                       | 0.183***  | 0.183***  | 0.205***  | 0.183***  |
|                                 | (0.006)   | (0.006)   | (0.006)   | (0.006)   |
| Self-employed                   | -0.061*** | -0.061*** | -0.044*** | -0.061*** |
|                                 | (0.009)   | (0.009)   | (0.010)   | (0.009)   |
| No job                          | 0.804***  | 0.804***  | 0.876***  | 0.804***  |
|                                 | (0.008)   | (0.008)   | (0.007)   | (0.008)   |
| Homemaker                       | 0.036***  | 0.036***  | 0.069***  | 0.036***  |
|                                 | (0.012)   | (0.012)   | (0.014)   | (0.012)   |
| Student                         | 0.140***  | 0.140***  | 0.168***  | 0.140***  |
|                                 | (0.008)   | (0.008)   | (0.007)   | (0.008)   |
| Retired                         | 0.003     | 0.003     | 0.037***  | 0.003     |
|                                 | (0.007)   | (0.007)   | (0.008)   | (0.007)   |
| Percent Black                   | 0.614     | 0.777     | -0.144    | 0.695     |
|                                 | (0.713)   | (0.754)   | (0.643)   | (0.791)   |
| Percent Hispanic                | -1.750*** | -1.638**  | -1.722**  | -1.593**  |
|                                 | (0.662)   | (0.665)   | (0.752)   | (0.683)   |
| Percent no access to internet   | 0.453**   | 0.455**   | 0.070     | 0.438**   |
|                                 | (0.186)   | (0.192)   | (0.145)   | (0.194)   |
| Percent unemployed              | 0.482     | 0.422     | 0.148     | 0.417     |
|                                 | (0.465)   | (0.470)   | (0.458)   | (0.487)   |
| Percent uninsured               | 0.247     | 0.270     | 0.449*    | 0.296     |
|                                 | (0.262)   | (0.260)   | (0.253)   | (0.266)   |
| Telehealth payment parity       | -0.007    | -0.007    | -0.010    | -0.007    |
|                                 | (0.009)   | (0.009)   | (0.011)   | (0.009)   |
| Observations                    | 5,683,627 | 5,683,627 | 4,028,804 | 5,683,627 |
| Mean dependent variable         | 3.70      | 3.70      | 3.49      | 3.70      |

*Panel B: Ever Diagnosed with Depression*

|      |         |        |         |
|------|---------|--------|---------|
| IMLC | 0.018** | 0.014* | 0.018** |
|------|---------|--------|---------|

|                                 | 2011–2023 |           | 2011–2019 | 2011–2023 |
|---------------------------------|-----------|-----------|-----------|-----------|
|                                 | (1)       | (2)       | (3)       | (4)       |
|                                 | (0.009)   |           | (0.007)   | (0.009)   |
|                                 | [0.190]   |           | [0.141]   | [0.194]   |
| PSYPACT                         | 0.003     |           |           |           |
|                                 | (0.010)   |           |           |           |
|                                 | [0.029]   |           |           |           |
| Either compact                  |           | 0.019**   |           |           |
|                                 |           | (0.008)   |           |           |
|                                 |           | [0.196]   |           |           |
| Male                            | -0.005    | -0.005    | 0.003     | -0.005    |
|                                 | (0.004)   | (0.004)   | (0.004)   | (0.004)   |
| Race and ethnicity              |           |           |           |           |
| Black                           | -0.005    | -0.005    | -0.013    | -0.005    |
|                                 | (0.007)   | (0.007)   | (0.009)   | (0.007)   |
| Other                           | 0.074***  | 0.074***  | 0.083***  | 0.074***  |
|                                 | (0.006)   | (0.006)   | (0.007)   | (0.006)   |
| Hispanic                        | -0.043*** | -0.043*** | -0.047*** | -0.043*** |
|                                 | (0.009)   | (0.009)   | (0.010)   | (0.009)   |
| Uninsured                       | 0.053***  | 0.053***  | 0.056***  | 0.053***  |
|                                 | (0.006)   | (0.006)   | (0.006)   | (0.006)   |
| Married                         | -0.029*** | -0.029*** | -0.010    | -0.029*** |
|                                 | (0.006)   | (0.006)   | (0.007)   | (0.006)   |
| Divorced                        | 0.063***  | 0.063***  | 0.071***  | 0.063***  |
|                                 | (0.003)   | (0.003)   | (0.004)   | (0.003)   |
| Widowed                         | 0.037***  | 0.037***  | 0.041***  | 0.037***  |
|                                 | (0.005)   | (0.005)   | (0.006)   | (0.005)   |
| Age                             |           |           |           |           |
| 35–44                           | -0.053*** | -0.053*** | -0.033*** | -0.053*** |
|                                 | (0.005)   | (0.005)   | (0.005)   | (0.005)   |
| 45–54                           | -0.098*** | -0.098*** | -0.067*** | -0.098*** |
|                                 | (0.006)   | (0.006)   | (0.006)   | (0.006)   |
| 55–64                           | -0.210*** | -0.210*** | -0.174*** | -0.210*** |
|                                 | (0.008)   | (0.008)   | (0.007)   | (0.008)   |
| 65+                             | -0.393*** | -0.393*** | -0.366*** | -0.393*** |
|                                 | (0.009)   | (0.009)   | (0.009)   | (0.009)   |
| High school or less             | -0.003    | -0.003    | 0.016     | -0.003    |
|                                 | (0.019)   | (0.019)   | (0.022)   | (0.019)   |
| Some college/Associate's degree | -0.017    | -0.017    | 0.000     | -0.017    |
|                                 | (0.019)   | (0.019)   | (0.022)   | (0.019)   |
| Bachelor's degree or higher     | -0.159*** | -0.159*** | -0.145*** | -0.159*** |
|                                 | (0.019)   | (0.019)   | (0.022)   | (0.019)   |
| Income                          |           |           |           |           |
| \$15,000–25,000                 | 0.380***  | 0.380***  | 0.417***  | 0.380***  |

|                               | 2011–2023 |           | 2011–2019 | 2011–2023 |
|-------------------------------|-----------|-----------|-----------|-----------|
|                               | (1)       | (2)       | (3)       | (4)       |
|                               | (0.008)   | (0.008)   | (0.010)   | (0.008)   |
| \$25,000–35,000               | 0.326***  | 0.326***  | 0.366***  | 0.326***  |
|                               | (0.008)   | (0.008)   | (0.009)   | (0.008)   |
| \$35,000–50,000               | 0.256***  | 0.256***  | 0.287***  | 0.256***  |
|                               | (0.007)   | (0.007)   | (0.009)   | (0.007)   |
| \$50,000–75,000               | 0.207***  | 0.207***  | 0.231***  | 0.207***  |
|                               | (0.006)   | (0.006)   | (0.007)   | (0.006)   |
| \$75,000+                     | 0.136***  | 0.136***  | 0.153***  | 0.136***  |
|                               | (0.005)   | (0.005)   | (0.006)   | (0.005)   |
| Self-employed                 | -0.022*** | -0.022*** | -0.015**  | -0.022*** |
|                               | (0.005)   | (0.005)   | (0.006)   | (0.005)   |
| No job                        | 0.432***  | 0.432***  | 0.480***  | 0.432***  |
|                               | (0.005)   | (0.005)   | (0.005)   | (0.005)   |
| Homemaker                     | 0.062***  | 0.062***  | 0.084***  | 0.062***  |
|                               | (0.009)   | (0.009)   | (0.011)   | (0.009)   |
| Student                       | 0.097***  | 0.097***  | 0.119***  | 0.097***  |
|                               | (0.006)   | (0.006)   | (0.006)   | (0.006)   |
| Retired                       | 0.050***  | 0.050***  | 0.082***  | 0.050***  |
|                               | (0.005)   | (0.005)   | (0.006)   | (0.005)   |
| Percent Black                 | 0.425     | 0.543     | 1.832***  | 0.434     |
|                               | (0.351)   | (0.349)   | (0.493)   | (0.350)   |
| Percent Hispanic              | -0.181    | -0.259    | -0.028    | -0.168    |
|                               | (0.712)   | (0.701)   | (0.874)   | (0.726)   |
| Percent no access to internet | 0.274**   | 0.289**   | 0.073     | 0.273**   |
|                               | (0.134)   | (0.131)   | (0.138)   | (0.135)   |
| Percent unemployed            | 0.589     | 0.591     | 0.558     | 0.582     |
|                               | (0.487)   | (0.482)   | (0.403)   | (0.492)   |
| Percent uninsured             | 0.356     | 0.345     | 0.493**   | 0.361*    |
|                               | (0.217)   | (0.211)   | (0.220)   | (0.215)   |
| Telehealth payment parity     | -0.002    | -0.001    | 0.002     | -0.002    |
|                               | (0.009)   | (0.009)   | (0.009)   | (0.009)   |
| Observations                  | 1,077,112 | 1,077,112 | 749,282   | 1,077,112 |
| Mean dependent variable       | 10.58     | 10.58     | 10.33     | 10.58     |

*Notes:* Estimates in brackets show marginal effects. Standard errors are clustered by state. All models include state and year fixed effects. \*\*\* $p < 0.01$ , \*\* $p < 0.05$ , \* $p < 0.10$ .

**Table A3:** Poor Mental Health Days and Compact Participation: Wooldridge DID

|                                 | IMLC  | PSYPACT | Either Compact |
|---------------------------------|-------|---------|----------------|
|                                 | (1)   | (2)     | (3)            |
| <i>Panel A: All Respondents</i> |       |         |                |
| IMLC participation              | 0.011 |         |                |

|                                                | IMLC<br>(1)       | PSYPACT<br>(2)      | Either Compact<br>(3) |
|------------------------------------------------|-------------------|---------------------|-----------------------|
|                                                | (0.014)           |                     |                       |
| PSYPACT participation                          |                   | 0.129***<br>(0.003) |                       |
| Either compact                                 |                   |                     | 0.011***<br>(0.003)   |
| Observations                                   | 306,062           | 331,461             | 374,209               |
| Mean dependent variable                        | 3.71              | 4.17                | 3.78                  |
| <i>Panel B: Ever Diagnosed with Depression</i> |                   |                     |                       |
| IMLC participation                             | -0.001<br>(0.013) |                     |                       |
| PSYPACT participation                          |                   | 0.077***<br>(0.017) |                       |
| Either compact                                 |                   |                     | -0.001<br>(0.012)     |
| Observations                                   | 60,997            | 65,879              | 73,678                |
| Mean dependent variable                        | 10.53             | 11.09               | 10.66                 |

*Notes:* Standard errors are clustered by state. All models include state and year fixed effects. \*\*\* $p < 0.01$ , \*\* $p < 0.05$ , \* $p < 0.10$ .

**Table A4:** Major Depressive Disorder and Compact Participation

|                                 | Major Depressive Disorder ( $\geq 14$ days) |                      |                      |                      | Major Depressive Disorder ( $\geq 21$ days) |                      |                      |                      |
|---------------------------------|---------------------------------------------|----------------------|----------------------|----------------------|---------------------------------------------|----------------------|----------------------|----------------------|
|                                 | 2011–2023                                   |                      | 2011–2019            | 2011–2023            | 2011–2023                                   |                      | 2011–2019            | 2011–2023            |
|                                 | (1)                                         | (2)                  | (3)                  | (4)                  | (5)                                         | (6)                  | (7)                  | (8)                  |
| <i>Panel A: All Respondents</i> |                                             |                      |                      |                      |                                             |                      |                      |                      |
| IMLC                            | 0.001<br>(0.001)                            |                      | 0.001<br>(0.001)     | 0.002<br>(0.001)     | 0.000<br>(0.001)                            |                      | 0.000<br>(0.001)     | 0.000<br>(0.001)     |
| PSYPACT                         | 0.004**<br>(0.002)                          |                      |                      |                      | 0.002**<br>(0.001)                          |                      |                      |                      |
| Either compact                  |                                             | 0.002*<br>(0.001)    |                      |                      |                                             | 0.000<br>(0.001)     |                      |                      |
| Male                            | -0.032***<br>(0.001)                        | -0.032***<br>(0.001) | -0.027***<br>(0.001) | -0.032***<br>(0.001) | -0.013***<br>(0.000)                        | -0.013***<br>(0.000) | -0.011***<br>(0.000) | -0.013***<br>(0.000) |
| Race and ethnicity              |                                             |                      |                      |                      |                                             |                      |                      |                      |
| Black                           | -0.036***<br>(0.002)                        | -0.036***<br>(0.002) | -0.035***<br>(0.003) | -0.036***<br>(0.002) | -0.024***<br>(0.002)                        | -0.024***<br>(0.002) | -0.024***<br>(0.002) | -0.024***<br>(0.002) |
| Other                           | -0.002<br>(0.003)                           | -0.002<br>(0.003)    | 0.000<br>(0.003)     | -0.002<br>(0.003)    | 0.002<br>(0.002)                            | 0.002<br>(0.002)     | 0.003<br>(0.002)     | 0.002<br>(0.002)     |
| Hispanic                        | -0.033***<br>(0.003)                        | -0.033***<br>(0.003) | -0.029***<br>(0.003) | -0.033***<br>(0.003) | -0.022***<br>(0.002)                        | -0.022***<br>(0.002) | -0.020***<br>(0.002) | -0.022***<br>(0.002) |
| Uninsured                       | 0.000<br>(0.002)                            | 0.000<br>(0.002)     | 0.003<br>(0.002)     | 0.000<br>(0.002)     | 0.004**<br>(0.002)                          | 0.004**<br>(0.002)   | 0.004**<br>(0.002)   | 0.004**<br>(0.002)   |
| Married                         | -0.022***<br>(0.002)                        | -0.022***<br>(0.002) | -0.014***<br>(0.002) | -0.022***<br>(0.002) | -0.010***<br>(0.001)                        | -0.010***<br>(0.001) | -0.005***<br>(0.001) | -0.010***<br>(0.001) |
| Divorced                        | 0.023***<br>(0.002)                         | 0.023***<br>(0.002)  | 0.027***<br>(0.002)  | 0.023***<br>(0.002)  | 0.020***<br>(0.001)                         | 0.020***<br>(0.001)  | 0.023***<br>(0.001)  | 0.020***<br>(0.001)  |
| Widowed                         | -0.010***<br>(0.002)                        | -0.010***<br>(0.002) | -0.005***<br>(0.002) | -0.010***<br>(0.002) | 0.001<br>(0.001)                            | 0.001<br>(0.001)     | 0.004***<br>(0.001)  | 0.001<br>(0.001)     |
| Age                             |                                             |                      |                      |                      |                                             |                      |                      |                      |
| 35–44                           | -0.009**<br>(0.001)                         | -0.009***<br>(0.001) | -0.001<br>(0.001)    | -0.009***<br>(0.001) | 0.003***<br>(0.001)                         | 0.003***<br>(0.001)  | 0.005***<br>(0.001)  | 0.003***<br>(0.001)  |

|                                 | Major Depressive Disorder ( $\geq 14$ days) |                      |                      |                      | Major Depressive Disorder ( $\geq 21$ days) |                      |                      |                      |
|---------------------------------|---------------------------------------------|----------------------|----------------------|----------------------|---------------------------------------------|----------------------|----------------------|----------------------|
|                                 | 2011–2023                                   |                      | 2011–2019            | 2011–2023            | 2011–2023                                   |                      | 2011–2019            | 2011–2023            |
|                                 | (1)                                         | (2)                  | (3)                  | (4)                  | (5)                                         | (6)                  | (7)                  | (8)                  |
| 45–54                           | -0.020***<br>(0.002)                        | -0.020***<br>(0.002) | -0.007***<br>(0.002) | -0.020***<br>(0.002) | -0.001<br>(0.001)                           | -0.001<br>(0.001)    | 0.004***<br>(0.001)  | -0.001<br>(0.001)    |
| 55–64                           | -0.047***<br>(0.002)                        | -0.047***<br>(0.002) | -0.031***<br>(0.002) | -0.047***<br>(0.002) | -0.016***<br>(0.001)                        | -0.016***<br>(0.001) | -0.010***<br>(0.001) | -0.016***<br>(0.001) |
| 65+                             | -0.087***<br>(0.003)                        | -0.087***<br>(0.003) | -0.073***<br>(0.003) | -0.087***<br>(0.003) | -0.041***<br>(0.002)                        | -0.041***<br>(0.002) | -0.035***<br>(0.002) | -0.041***<br>(0.002) |
| High school or less             | 0.011***<br>(0.002)                         | 0.011***<br>(0.002)  | 0.009***<br>(0.002)  | 0.011***<br>(0.002)  | 0.006***<br>(0.002)                         | 0.006***<br>(0.002)  | 0.006***<br>(0.002)  | 0.006***<br>(0.002)  |
| Some college/Associate's degree | 0.015***<br>(0.002)                         | 0.015***<br>(0.002)  | 0.012***<br>(0.002)  | 0.015***<br>(0.002)  | 0.007***<br>(0.002)                         | 0.007***<br>(0.002)  | 0.006***<br>(0.002)  | 0.007***<br>(0.002)  |
| Bachelor's degree or higher     | -0.010***<br>(0.002)                        | -0.010***<br>(0.002) | -0.011***<br>(0.002) | -0.010***<br>(0.002) | -0.009***<br>(0.002)                        | -0.009***<br>(0.002) | -0.009***<br>(0.002) | -0.009***<br>(0.002) |
| Income                          |                                             |                      |                      |                      |                                             |                      |                      |                      |
| Less than \$15,000              | 0.100***<br>(0.002)                         | 0.100***<br>(0.002)  | 0.099***<br>(0.002)  | 0.100***<br>(0.002)  | 0.069***<br>(0.002)                         | 0.069***<br>(0.002)  | 0.068***<br>(0.002)  | 0.069***<br>(0.002)  |
| \$15,000–25,000                 | 0.062***<br>(0.002)                         | 0.062***<br>(0.002)  | 0.060***<br>(0.001)  | 0.062***<br>(0.002)  | 0.038***<br>(0.001)                         | 0.038***<br>(0.001)  | 0.037***<br>(0.001)  | 0.038***<br>(0.001)  |
| \$25,000–35,000                 | 0.041***<br>(0.001)                         | 0.041***<br>(0.001)  | 0.038***<br>(0.001)  | 0.041***<br>(0.001)  | 0.024***<br>(0.001)                         | 0.024***<br>(0.001)  | 0.022***<br>(0.001)  | 0.024***<br>(0.001)  |
| \$35,000–50,000                 | 0.029***<br>(0.001)                         | 0.029***<br>(0.001)  | 0.027***<br>(0.001)  | 0.029***<br>(0.001)  | 0.016***<br>(0.000)                         | 0.016***<br>(0.000)  | 0.015***<br>(0.001)  | 0.016***<br>(0.000)  |
| \$50,000–75,000                 | 0.018***<br>(0.001)                         | 0.017***<br>(0.001)  | 0.015***<br>(0.001)  | 0.017***<br>(0.001)  | 0.009***<br>(0.000)                         | 0.009***<br>(0.000)  | 0.009***<br>(0.000)  | 0.009***<br>(0.000)  |
| Self-employed                   | -0.001<br>(0.001)                           | -0.001<br>(0.001)    | 0.000<br>(0.001)     | -0.001<br>(0.001)    | 0.002***<br>(0.001)                         | 0.002***<br>(0.001)  | 0.003***<br>(0.001)  | 0.002***<br>(0.001)  |
| No job                          | 0.186***<br>(0.003)                         | 0.186***<br>(0.003)  | 0.192***<br>(0.003)  | 0.186***<br>(0.003)  | 0.132***<br>(0.002)                         | 0.132***<br>(0.002)  | 0.137***<br>(0.002)  | 0.132***<br>(0.002)  |
| Homemaker                       | 0.006***<br>(0.002)                         | 0.006***<br>(0.002)  | 0.009***<br>(0.002)  | 0.006***<br>(0.002)  | 0.006***<br>(0.001)                         | 0.006***<br>(0.001)  | 0.008***<br>(0.001)  | 0.006***<br>(0.001)  |
| Student                         | 0.015***                                    | 0.015***             | 0.016***             | 0.015***             | -0.003**                                    | -0.003**             | -0.001               | -0.003**             |

|                                                | Major Depressive Disorder ( $\geq 14$ days) |                      |                      |                      | Major Depressive Disorder ( $\geq 21$ days) |                      |                     |                      |
|------------------------------------------------|---------------------------------------------|----------------------|----------------------|----------------------|---------------------------------------------|----------------------|---------------------|----------------------|
|                                                | 2011–2023                                   |                      | 2011–2019            | 2011–2023            | 2011–2023                                   |                      | 2011–2019           | 2011–2023            |
|                                                | (1)                                         | (2)                  | (3)                  | (4)                  | (5)                                         | (6)                  | (7)                 | (8)                  |
| Retired                                        | (0.002)<br>0.012***                         | (0.002)<br>0.012***  | (0.001)<br>0.015***  | (0.002)<br>0.012***  | (0.001)<br>0.011***                         | (0.001)<br>0.011***  | (0.001)<br>0.013*** | (0.001)<br>0.011***  |
| Percent Black                                  | (0.001)<br>0.132*                           | (0.001)<br>0.150**   | (0.001)<br>0.050     | (0.001)<br>0.143*    | (0.001)<br>0.051                            | (0.001)<br>0.059     | (0.001)<br>-0.014   | (0.001)<br>0.058     |
| Percent Hispanic                               | (0.070)<br>-0.253***                        | (0.072)<br>-0.230*** | (0.083)<br>-0.239*** | (0.077)<br>-0.228*** | (0.044)<br>-0.174***                        | (0.045)<br>-0.160*** | (0.048)<br>-0.119*  | (0.046)<br>-0.160*** |
| Percent no access to internet                  | (0.082)<br>0.019                            | (0.083)<br>0.018     | (0.087)<br>-0.012    | (0.085)<br>0.017     | (0.058)<br>0.002                            | (0.058)<br>0.002     | (0.069)<br>-0.016   | (0.058)<br>0.001     |
| Percent unemployed                             | (0.024)<br>-0.003                           | (0.025)<br>-0.013    | (0.016)<br>-0.031    | (0.025)<br>-0.013    | (0.012)<br>0.003                            | (0.013)<br>-0.003    | (0.010)<br>-0.028   | (0.013)<br>-0.003    |
| Percent uninsured                              | (0.047)<br>-0.002                           | (0.048)<br>0.001     | (0.050)<br>0.031     | (0.050)<br>0.004     | (0.029)<br>-0.010                           | (0.030)<br>-0.007    | (0.031)<br>0.024    | (0.030)<br>-0.006    |
| Telehealth payment parity                      | (0.033)<br>-0.001                           | (0.033)<br>-0.001    | (0.033)<br>-0.002    | (0.034)<br>-0.001    | (0.021)<br>-0.001*                          | (0.021)<br>-0.001*   | (0.019)<br>-0.001   | (0.021)<br>-0.002*   |
|                                                | (0.001)                                     | (0.001)              | (0.001)              | (0.001)              | (0.001)                                     | (0.001)              | (0.001)             | (0.001)              |
| Observations                                   | 5,683,627                                   | 5,683,627            | 4,028,804            | 5,683,627            | 5,683,627                                   | 5,683,627            | 4,028,804           | 5,683,627            |
| R <sup>2</sup>                                 | 0.081                                       | 0.081                | 0.084                | 0.081                | 0.058                                       | 0.058                | 0.063               | 0.058                |
| Mean dependent variable                        | 0.115                                       | 0.115                | 0.109                | 0.115                | 0.066                                       | 0.066                | 0.064               | 0.066                |
| <i>Panel B: Ever Diagnosed with Depression</i> |                                             |                      |                      |                      |                                             |                      |                     |                      |
| IMLC                                           | 0.007**<br>(0.003)                          |                      | 0.006*<br>(0.003)    | 0.007**<br>(0.003)   | 0.002<br>(0.002)                            |                      | 0.000<br>(0.002)    | 0.002<br>(0.002)     |
| PSYPACT                                        | 0.001<br>(0.004)                            |                      |                      |                      | 0.001<br>(0.003)                            |                      |                     |                      |
| Either compact                                 |                                             | 0.007**<br>(0.003)   |                      |                      |                                             | 0.002<br>(0.002)     |                     |                      |
| Male                                           | -0.003*<br>(0.002)                          | -0.003*<br>(0.002)   | 0.002<br>(0.002)     | -0.003*<br>(0.002)   | 0.011***<br>(0.001)                         | 0.011***<br>(0.001)  | 0.013***<br>(0.001) | 0.011***<br>(0.001)  |
| Race and ethnicity                             |                                             |                      |                      |                      |                                             |                      |                     |                      |

|                                 | Major Depressive Disorder ( $\geq 14$ days) |                      |                      |                      | Major Depressive Disorder ( $\geq 21$ days) |                      |                      |                      |
|---------------------------------|---------------------------------------------|----------------------|----------------------|----------------------|---------------------------------------------|----------------------|----------------------|----------------------|
|                                 | 2011–2023                                   |                      | 2011–2019            | 2011–2023            | 2011–2023                                   |                      | 2011–2019            | 2011–2023            |
|                                 | (1)                                         | (2)                  | (3)                  | (4)                  | (5)                                         | (6)                  | (7)                  | (8)                  |
| Black                           | -0.003<br>(0.003)                           | -0.003<br>(0.003)    | -0.006<br>(0.004)    | -0.003<br>(0.003)    | -0.011***<br>(0.004)                        | -0.011***<br>(0.004) | -0.014***<br>(0.004) | -0.011***<br>(0.004) |
| Other                           | 0.032***<br>(0.003)                         | 0.032***<br>(0.003)  | 0.036***<br>(0.003)  | 0.032***<br>(0.003)  | 0.029***<br>(0.003)                         | 0.029***<br>(0.003)  | 0.032***<br>(0.003)  | 0.029***<br>(0.003)  |
| Hispanic                        | -0.017***<br>(0.004)                        | -0.017***<br>(0.004) | -0.018***<br>(0.004) | -0.017***<br>(0.004) | -0.015***<br>(0.003)                        | -0.015***<br>(0.003) | -0.018***<br>(0.003) | -0.015***<br>(0.003) |
| Uninsured                       | 0.024***<br>(0.003)                         | 0.024***<br>(0.003)  | 0.027***<br>(0.003)  | 0.024***<br>(0.003)  | 0.028***<br>(0.003)                         | 0.028***<br>(0.003)  | 0.027***<br>(0.003)  | 0.028***<br>(0.003)  |
| Married                         | -0.012***<br>(0.003)                        | -0.012***<br>(0.003) | -0.003<br>(0.003)    | -0.012***<br>(0.003) | -0.001<br>(0.002)                           | -0.001<br>(0.002)    | 0.005**<br>(0.002)   | -0.001<br>(0.002)    |
| Divorced                        | 0.025***<br>(0.002)                         | 0.025***<br>(0.002)  | 0.029***<br>(0.002)  | 0.025***<br>(0.002)  | 0.028***<br>(0.001)                         | 0.028***<br>(0.001)  | 0.032***<br>(0.002)  | 0.028***<br>(0.001)  |
| Widowed                         | 0.006**<br>(0.002)                          | 0.006**<br>(0.002)   | 0.010***<br>(0.003)  | 0.006**<br>(0.002)   | 0.018***<br>(0.002)                         | 0.018***<br>(0.002)  | 0.022***<br>(0.002)  | 0.018***<br>(0.002)  |
| Age                             |                                             |                      |                      |                      |                                             |                      |                      |                      |
| 35–44                           | -0.028***<br>(0.002)                        | -0.028***<br>(0.002) | -0.016***<br>(0.002) | -0.028***<br>(0.002) | 0.000<br>(0.002)                            | 0.000<br>(0.002)     | 0.003<br>(0.002)     | 0.000<br>(0.002)     |
| 45–54                           | -0.045***<br>(0.003)                        | -0.045***<br>(0.003) | -0.027***<br>(0.003) | -0.045***<br>(0.003) | -0.005***<br>(0.002)                        | -0.005***<br>(0.002) | 0.002<br>(0.002)     | -0.005***<br>(0.002) |
| 55–64                           | -0.093***<br>(0.003)                        | -0.093***<br>(0.003) | -0.073***<br>(0.003) | -0.093***<br>(0.003) | -0.035***<br>(0.002)                        | -0.035***<br>(0.002) | -0.028***<br>(0.003) | -0.035***<br>(0.002) |
| 65+                             | -0.157***<br>(0.003)                        | -0.157***<br>(0.003) | -0.138***<br>(0.004) | -0.157***<br>(0.003) | -0.077***<br>(0.003)                        | -0.077***<br>(0.003) | -0.073***<br>(0.003) | -0.077***<br>(0.003) |
| High school or less             | 0.002<br>(0.009)                            | 0.002<br>(0.009)     | 0.008<br>(0.010)     | 0.002<br>(0.009)     | -0.007<br>(0.010)                           | -0.007<br>(0.010)    | 0.000<br>(0.012)     | -0.007<br>(0.010)    |
| Some college/Associate's degree | -0.007<br>(0.009)                           | -0.007<br>(0.009)    | -0.002<br>(0.009)    | -0.007<br>(0.009)    | -0.017<br>(0.010)                           | -0.017<br>(0.010)    | -0.010<br>(0.012)    | -0.017<br>(0.010)    |
| Bachelor's degree or higher     | -0.063***<br>(0.009)                        | -0.063***<br>(0.009) | -0.056***<br>(0.010) | -0.063***<br>(0.009) | -0.056***<br>(0.010)                        | -0.056***<br>(0.010) | -0.045***<br>(0.012) | -0.056***<br>(0.010) |
| Income                          |                                             |                      |                      |                      |                                             |                      |                      |                      |

|                               | Major Depressive Disorder ( $\geq 14$ days) |                     |                     |                     | Major Depressive Disorder ( $\geq 21$ days) |                     |                     |                     |
|-------------------------------|---------------------------------------------|---------------------|---------------------|---------------------|---------------------------------------------|---------------------|---------------------|---------------------|
|                               | 2011–2023                                   |                     | 2011–2019           | 2011–2023           | 2011–2023                                   |                     | 2011–2019           | 2011–2023           |
|                               | (1)                                         | (2)                 | (3)                 | (4)                 | (5)                                         | (6)                 | (7)                 | (8)                 |
| Less than \$15,000            | 0.153***<br>(0.003)                         | 0.153***<br>(0.003) | 0.155***<br>(0.003) | 0.153***<br>(0.003) | 0.117***<br>(0.002)                         | 0.117***<br>(0.002) | 0.116***<br>(0.003) | 0.117***<br>(0.002) |
| \$15,000–25,000               | 0.120***<br>(0.002)                         | 0.120***<br>(0.002) | 0.122***<br>(0.002) | 0.120***<br>(0.002) | 0.080***<br>(0.002)                         | 0.080***<br>(0.002) | 0.081***<br>(0.002) | 0.080***<br>(0.002) |
| \$25,000–35,000               | 0.087***<br>(0.002)                         | 0.087***<br>(0.002) | 0.085***<br>(0.002) | 0.087***<br>(0.002) | 0.057***<br>(0.002)                         | 0.057***<br>(0.002) | 0.054***<br>(0.002) | 0.057***<br>(0.002) |
| \$35,000–50,000               | 0.068***<br>(0.002)                         | 0.068***<br>(0.002) | 0.066***<br>(0.002) | 0.068***<br>(0.002) | 0.041***<br>(0.002)                         | 0.041***<br>(0.002) | 0.039***<br>(0.002) | 0.041***<br>(0.002) |
| \$50,000–75,000               | 0.043***<br>(0.002)                         | 0.043***<br>(0.002) | 0.041***<br>(0.002) | 0.043***<br>(0.002) | 0.025***<br>(0.001)                         | 0.025***<br>(0.001) | 0.025***<br>(0.001) | 0.025***<br>(0.001) |
| Self-employed                 | -0.004**<br>(0.002)                         | -0.004**<br>(0.002) | -0.001<br>(0.002)   | -0.004**<br>(0.002) | 0.004**<br>(0.002)                          | 0.004**<br>(0.002)  | 0.004**<br>(0.002)  | 0.004**<br>(0.002)  |
| No job                        | 0.210***<br>(0.002)                         | 0.210***<br>(0.002) | 0.224***<br>(0.002) | 0.210***<br>(0.002) | 0.170***<br>(0.002)                         | 0.170***<br>(0.002) | 0.180***<br>(0.002) | 0.170***<br>(0.002) |
| Homemaker                     | 0.025***<br>(0.004)                         | 0.025***<br>(0.004) | 0.032***<br>(0.004) | 0.025***<br>(0.004) | 0.021***<br>(0.003)                         | 0.021***<br>(0.003) | 0.025***<br>(0.003) | 0.021***<br>(0.003) |
| Student                       | 0.041***<br>(0.003)                         | 0.041***<br>(0.003) | 0.042***<br>(0.003) | 0.041***<br>(0.003) | 0.003<br>(0.003)                            | 0.003<br>(0.003)    | 0.007***<br>(0.003) | 0.003<br>(0.003)    |
| Retired                       | 0.032***<br>(0.002)                         | 0.032***<br>(0.002) | 0.041***<br>(0.002) | 0.032***<br>(0.002) | 0.032***<br>(0.002)                         | 0.032***<br>(0.002) | 0.038***<br>(0.002) | 0.032***<br>(0.002) |
| Percent Black                 | 0.385**<br>(0.173)                          | 0.434**<br>(0.171)  | 0.997***<br>(0.213) | 0.389**<br>(0.171)  | 0.104<br>(0.154)                            | 0.118<br>(0.153)    | 0.441***<br>(0.159) | 0.107<br>(0.153)    |
| Percent Hispanic              | -0.071<br>(0.273)                           | -0.106<br>(0.271)   | -0.054<br>(0.338)   | -0.065<br>(0.279)   | -0.043<br>(0.195)                           | -0.044<br>(0.191)   | 0.217<br>(0.193)    | -0.037<br>(0.198)   |
| Percent no access to internet | 0.066<br>(0.049)                            | 0.072<br>(0.048)    | -0.001<br>(0.057)   | 0.066<br>(0.049)    | 0.031<br>(0.035)                            | 0.032<br>(0.035)    | -0.008<br>(0.042)   | 0.030<br>(0.036)    |
| Percent unemployed            | 0.113<br>(0.178)                            | 0.115<br>(0.177)    | 0.091<br>(0.157)    | 0.110<br>(0.180)    | 0.122<br>(0.112)                            | 0.120<br>(0.113)    | 0.044<br>(0.098)    | 0.119<br>(0.114)    |
| Percent uninsured             | 0.093<br>(0.086)                            | 0.090<br>(0.083)    | 0.153<br>(0.092)    | 0.095<br>(0.085)    | 0.074<br>(0.063)                            | 0.074<br>(0.062)    | 0.148**<br>(0.067)  | 0.076<br>(0.062)    |

|                           | Major Depressive Disorder ( $\geq 14$ days) |                   |                   |                   | Major Depressive Disorder ( $\geq 21$ days) |                   |                  |                   |
|---------------------------|---------------------------------------------|-------------------|-------------------|-------------------|---------------------------------------------|-------------------|------------------|-------------------|
|                           | 2011–2023                                   |                   | 2011–2019         | 2011–2023         | 2011–2023                                   |                   | 2011–2019        | 2011–2023         |
|                           | (1)                                         | (2)               | (3)               | (4)               | (5)                                         | (6)               | (7)              | (8)               |
| Telehealth payment parity | -0.001<br>(0.003)                           | -0.001<br>(0.004) | -0.001<br>(0.004) | -0.001<br>(0.003) | -0.003<br>(0.002)                           | -0.003<br>(0.002) | 0.000<br>(0.003) | -0.003<br>(0.002) |
| Observations              | 1,077,112                                   | 1,077,112         | 749,282           | 1,077,112         | 1,077,112                                   | 1,077,112         | 749,282          | 1,077,112         |
| R <sup>2</sup>            | 0.097                                       | 0.097             | 0.105             | 0.097             | 0.075                                       | 0.075             | 0.082            | 0.075             |
| Mean dependent variable   | 0.360                                       | 0.360             | 0.350             | 0.360             | 0.220                                       | 0.220             | 0.218            | 0.220             |

*Notes:* Standard errors are clustered by state. All models include state and year fixed effects. \*\*\* $p < 0.01$ , \*\* $p < 0.05$ , \* $p < 0.10$ .

**Table A5:** State-level Mental Health and Compact Participation

|                               | Suicide Rate per 100,000 |                        |                       |                        | SSRI Prescribing Rate |                   |
|-------------------------------|--------------------------|------------------------|-----------------------|------------------------|-----------------------|-------------------|
|                               | 2011–2023                |                        | 2011–2019             | 2011–2023              | 2013–2019             | 2013–2022         |
|                               | (1)                      | (2)                    | (3)                   | (4)                    | (5)                   | (6)               |
| IMLC                          | 0.502**<br>(0.215)       |                        | 0.155<br>(0.198)      | 0.449**<br>(0.216)     | 0<br>(0.000)          | 0<br>(0.000)      |
| PSYPACT                       | -0.388*<br>(0.228)       |                        |                       |                        |                       |                   |
| Either compact                |                          | 0.329<br>(0.210)       |                       |                        |                       |                   |
| Percent Black                 | 6.760<br>(15.241)        | 10.693<br>(15.445)     | 7.252<br>(16.426)     | 7.461<br>(15.630)      | 0.052**<br>(0.021)    | 0.026*<br>(0.013) |
| Percent Hispanic              | -49.637**<br>(19.055)    | -54.929***<br>(18.441) | -46.796**<br>(19.854) | -51.363***<br>(18.747) | -0.004<br>(0.024)     | -0.007<br>(0.036) |
| Percent no access to internet | -0.511<br>(4.423)        | -0.354<br>(4.391)      | 7.400**<br>(3.400)    | -0.460<br>(4.487)      | -0.011<br>(0.009)     | -0.010<br>(0.012) |
| Percent unemployed            | 21.107**<br>(8.863)      | 21.866**<br>(8.728)    | 36.497***<br>(9.314)  | 21.775**<br>(8.729)    | -0.001<br>(0.007)     | 0.006<br>(0.006)  |
| Percent uninsured             | -4.607<br>(8.903)        | -5.234<br>(8.914)      | -9.380<br>(6.603)     | -5.330<br>(8.979)      | 0.008<br>(0.006)      | 0.014<br>(0.009)  |
| Telehealth payment parity     | -0.270<br>(0.256)        | -0.283<br>(0.265)      | -0.397<br>(0.363)     | -0.277<br>(0.262)      | 0<br>(0.000)          | 0<br>(0.000)      |
| Observations                  | 663                      | 663                    | 459                   | 663                    | 357                   | 510               |
| R <sup>2</sup>                | 0.954                    | 0.954                  | 0.958                 | 0.954                  | 0.993                 | 0.987             |
| Mean dependent variable       | 15.62                    | 15.62                  | 15.28                 | 15.62                  | 0.031                 | 0.031             |

Notes: Standard errors are clustered by state. All models include state and year fixed effects. \*\*\* $p < 0.01$ , \*\* $p < 0.05$ , \* $p < 0.10$ .

**Table A6:** Poor Mental Health Days and Compact Participation

|                                 | (1)                           | (2)                           |
|---------------------------------|-------------------------------|-------------------------------|
| <i>Panel A: All Respondents</i> |                               |                               |
| IMLC                            | 0.012<br>(0.010)<br>[0.044]   |                               |
| PSYPACT                         | 0.028**<br>(0.011)<br>[0.103] |                               |
| Either compact                  |                               | 0.021**<br>(0.009)<br>[0.077] |
| Male                            | -0.289***                     | -0.289***                     |

|                                 | (1)                  | (2)                  |
|---------------------------------|----------------------|----------------------|
|                                 | (0.007)              | (0.007)              |
| Race and ethnicity              |                      |                      |
| Black                           | -0.236***<br>(0.013) | -0.236***<br>(0.013) |
| Other                           | -0.026<br>(0.018)    | -0.026<br>(0.018)    |
| Hispanic                        | -0.240***<br>(0.018) | -0.240***<br>(0.018) |
| Uninsured                       | -0.003<br>(0.011)    | -0.003<br>(0.011)    |
| Married                         | -0.196***<br>(0.012) | -0.196***<br>(0.012) |
| Divorced                        | 0.131***<br>(0.006)  | 0.131***<br>(0.006)  |
| Widowed                         | -0.005<br>(0.009)    | -0.005<br>(0.009)    |
| Age                             |                      |                      |
| 35–44                           | -0.064***<br>(0.006) | -0.064***<br>(0.006) |
| 45–54                           | -0.163***<br>(0.010) | -0.163***<br>(0.010) |
| 55–64                           | -0.353***<br>(0.013) | -0.353***<br>(0.013) |
| 65+                             | -0.741***<br>(0.016) | -0.741***<br>(0.016) |
| High school or less             | 0.082***<br>(0.017)  | 0.082***<br>(0.017)  |
| Some college/Associate's degree | 0.117***<br>(0.017)  | 0.116***<br>(0.017)  |
| Bachelor's degree or higher     | -0.067***<br>(0.019) | -0.067***<br>(0.019) |
| Income                          |                      |                      |
| \$15,000–25,000                 | 0.580***<br>(0.014)  | 0.580***<br>(0.014)  |
| \$25,000–35,000                 | 0.472***<br>(0.014)  | 0.472***<br>(0.014)  |
| \$35,000–50,000                 | 0.357***<br>(0.012)  | 0.357***<br>(0.012)  |
| \$50,000–75,000                 | 0.277***<br>(0.009)  | 0.277***<br>(0.009)  |
| \$75,000+                       | 0.183***<br>(0.006)  | 0.183***<br>(0.006)  |
| Self-employed                   | -0.061***<br>(0.009) | -0.061***<br>(0.009) |

|                                                | (1)                           | (2)                           |
|------------------------------------------------|-------------------------------|-------------------------------|
| No job                                         | 0.804***<br>(0.008)           | 0.804***<br>(0.008)           |
| Homemaker                                      | 0.036***<br>(0.012)           | 0.036***<br>(0.012)           |
| Student                                        | 0.140***<br>(0.008)           | 0.140***<br>(0.008)           |
| Retired                                        | 0.003<br>(0.007)              | 0.003<br>(0.007)              |
| Percent Black                                  | 0.589<br>(0.704)              | 0.748<br>(0.749)              |
| Percent Hispanic                               | -1.748***<br>(0.646)          | -1.617**<br>(0.656)           |
| Percent no access to internet                  | 0.460**<br>(0.186)            | 0.458**<br>(0.193)            |
| Percent unemployed                             | 0.493<br>(0.458)              | 0.43<br>(0.466)               |
| Percent uninsured                              | 0.225<br>(0.255)              | 0.248<br>(0.254)              |
| Telehealth payment parity                      | -0.006<br>(0.009)             | -0.006<br>(0.009)             |
| Emergency expansions                           |                               |                               |
| Out-of-state license endorsement               | -0.005<br>(0.009)             | -0.005<br>(0.010)             |
| Telehealth                                     | 0.017<br>(0.023)              | 0.014<br>(0.023)              |
| Observations                                   | 5,683,627                     | 5,683,627                     |
| Mean dependent variable                        | 3.70                          | 3.70                          |
| <i>Panel B: Ever Diagnosed with Depression</i> |                               |                               |
| IMLC                                           | 0.017**<br>(0.008)<br>[0.185] |                               |
| PSYPACT                                        | 0.004<br>(0.010)<br>[0.044]   |                               |
| Either compact                                 |                               | 0.019**<br>(0.008)<br>[0.199] |
| Male                                           | -0.005<br>(0.004)             | -0.005<br>(0.004)             |
| Race and ethnicity                             |                               |                               |
| Black                                          | -0.005<br>(0.007)             | -0.005<br>(0.007)             |
| Other                                          | 0.074***                      | 0.074***                      |

|                                 | (1)       | (2)       |
|---------------------------------|-----------|-----------|
|                                 | (0.006)   | (0.006)   |
| Hispanic                        | -0.043*** | -0.043*** |
|                                 | (0.009)   | (0.009)   |
| Uninsured                       | 0.053***  | 0.053***  |
|                                 | (0.006)   | (0.006)   |
| Married                         | -0.029*** | -0.029*** |
|                                 | (0.006)   | (0.006)   |
| Divorced                        | 0.063***  | 0.063***  |
|                                 | (0.003)   | (0.003)   |
| Widowed                         | 0.037***  | 0.037***  |
|                                 | (0.005)   | (0.005)   |
| Age                             |           |           |
| 35–44                           | -0.053*** | -0.053*** |
|                                 | (0.005)   | (0.005)   |
| 45–54                           | -0.098*** | -0.098*** |
|                                 | (0.006)   | (0.006)   |
| 55–64                           | -0.209*** | -0.209*** |
|                                 | (0.008)   | (0.008)   |
| 65+                             | -0.393*** | -0.393*** |
|                                 | (0.009)   | (0.009)   |
| High school or less             | -0.003    | -0.003    |
|                                 | (0.019)   | (0.019)   |
| Some college/Associate's degree | -0.017    | -0.017    |
|                                 | (0.019)   | (0.019)   |
| Bachelor's degree or higher     | -0.159*** | -0.159*** |
|                                 | (0.019)   | (0.019)   |
| Income                          |           |           |
| \$15,000–25,000                 | 0.380***  | 0.380***  |
|                                 | (0.008)   | (0.008)   |
| \$25,000–35,000                 | 0.326***  | 0.326***  |
|                                 | (0.008)   | (0.008)   |
| \$35,000–50,000                 | 0.256***  | 0.256***  |
|                                 | (0.007)   | (0.007)   |
| \$50,000–75,000                 | 0.207***  | 0.207***  |
|                                 | (0.006)   | (0.006)   |
| \$75,000+                       | 0.136***  | 0.136***  |
|                                 | (0.005)   | (0.005)   |
| Self-employed                   | -0.022*** | -0.022*** |
|                                 | (0.005)   | (0.005)   |
| No job                          | 0.432***  | 0.432***  |
|                                 | (0.005)   | (0.005)   |
| Homemaker                       | 0.062***  | 0.062***  |
|                                 | (0.009)   | (0.009)   |
| Student                         | 0.097***  | 0.097***  |

|                                  | (1)       | (2)       |
|----------------------------------|-----------|-----------|
|                                  | (0.006)   | (0.006)   |
| Retired                          | 0.050***  | 0.050***  |
|                                  | (0.005)   | (0.005)   |
| Percent Black                    | 0.423     | 0.541     |
|                                  | (0.349)   | (0.351)   |
| Percent Hispanic                 | -0.196    | -0.259    |
|                                  | (0.704)   | (0.698)   |
| Percent no access to internet    | 0.285**   | 0.299**   |
|                                  | (0.133)   | (0.130)   |
| Percent unemployed               | 0.589     | 0.588     |
|                                  | (0.479)   | (0.474)   |
| Percent uninsured                | 0.350*    | 0.339*    |
|                                  | (0.209)   | (0.202)   |
| Telehealth payment parity        | -0.002    | -0.001    |
|                                  | (0.009)   | (0.009)   |
| Emergency expansions             |           |           |
| Out-of-state license endorsement | 0.001     | 0.001     |
|                                  | (0.008)   | (0.008)   |
| Telehealth                       | 0.012     | 0.013     |
|                                  | (0.022)   | (0.022)   |
| Observations                     | 1,077,112 | 1,077,112 |
| Mean dependent variable          | 10.58     | 10.58     |

*Notes:* Estimates in brackets show marginal effects. Standard errors are clustered by state. All models include state and year fixed effects. All models include state and year fixed effects. Models additionally control for temporary COVID-19 emergency expansions permitting out-of-state license endorsement and telehealth practice. \*\*\* $p < 0.01$ , \*\* $p < 0.05$ , \* $p < 0.10$ .

**Table A7:** Major Depressive Disorder and Compact Participation

|                                 | MDD ( $\geq 14$ days) |           | MDD ( $\geq 21$ days) |           |
|---------------------------------|-----------------------|-----------|-----------------------|-----------|
|                                 | (1)                   | (2)       | (3)                   | (4)       |
| <i>Panel A: All Respondents</i> |                       |           |                       |           |
| IMLC                            | 0.001                 |           | 0.000                 |           |
|                                 | (0.001)               |           | (0.001)               |           |
| PSYPACT                         | 0.004***              |           | 0.002**               |           |
|                                 | (0.001)               |           | (0.001)               |           |
| Either compact                  |                       | 0.002**   |                       | 0.000     |
|                                 |                       | (0.001)   |                       | (0.001)   |
| Male                            | -0.032***             | -0.032*** | -0.013***             | -0.013*** |
|                                 | (0.001)               | (0.001)   | (0.000)               | (0.000)   |
| Race and ethnicity              |                       |           |                       |           |

|                                 |  | MDD ( $\geq 14$ days) |                      | MDD ( $\geq 21$ days) |                      |
|---------------------------------|--|-----------------------|----------------------|-----------------------|----------------------|
|                                 |  | (1)                   | (2)                  | (3)                   | (4)                  |
| Black                           |  | -0.036***<br>(0.002)  | -0.036***<br>(0.002) | -0.024***<br>(0.002)  | -0.024***<br>(0.002) |
| Other                           |  | -0.002<br>(0.003)     | -0.002<br>(0.003)    | 0.002<br>(0.002)      | 0.002<br>(0.002)     |
| Hispanic                        |  | -0.033***<br>(0.003)  | -0.033***<br>(0.003) | -0.022***<br>(0.002)  | -0.022***<br>(0.002) |
| Uninsured                       |  | 0<br>(0.002)          | 0<br>(0.002)         | 0.004**<br>(0.002)    | 0.004**<br>(0.002)   |
| Married                         |  | -0.022***<br>(0.002)  | -0.022***<br>(0.002) | -0.010***<br>(0.001)  | -0.010***<br>(0.001) |
| Divorced                        |  | 0.023***<br>(0.002)   | 0.023***<br>(0.002)  | 0.020***<br>(0.001)   | 0.020***<br>(0.001)  |
| Widowed                         |  | -0.010***<br>(0.002)  | -0.010***<br>(0.002) | 0.001<br>(0.001)      | 0.001<br>(0.001)     |
| Age                             |  |                       |                      |                       |                      |
| 35–44                           |  | -0.009***<br>(0.001)  | -0.009***<br>(0.001) | 0.003***<br>(0.001)   | 0.003***<br>(0.001)  |
| 45–54                           |  | -0.020***<br>(0.002)  | -0.020***<br>(0.002) | -0.001<br>(0.001)     | -0.001<br>(0.001)    |
| 55–64                           |  | -0.047***<br>(0.002)  | -0.047***<br>(0.002) | -0.016***<br>(0.001)  | -0.016***<br>(0.001) |
| 65+                             |  | -0.087***<br>(0.003)  | -0.087***<br>(0.003) | -0.041***<br>(0.002)  | -0.041***<br>(0.002) |
| High school or less             |  | 0.011***<br>(0.002)   | 0.011***<br>(0.002)  | 0.006***<br>(0.002)   | 0.006***<br>(0.002)  |
| Some college/Associate's degree |  | 0.015***<br>(0.002)   | 0.015***<br>(0.002)  | 0.007***<br>(0.002)   | 0.007***<br>(0.002)  |
| Bachelor's degree or higher     |  | -0.010***<br>(0.002)  | -0.010***<br>(0.002) | -0.009***<br>(0.002)  | -0.009***<br>(0.002) |
| Income                          |  |                       |                      |                       |                      |
| \$15,000–25,000                 |  | 0.100***<br>(0.002)   | 0.100***<br>(0.002)  | 0.069***<br>(0.002)   | 0.069***<br>(0.002)  |
| \$25,000–35,000                 |  | 0.062***<br>(0.002)   | 0.062***<br>(0.002)  | 0.038***<br>(0.001)   | 0.038***<br>(0.001)  |
| \$35,000–50,000                 |  | 0.041***<br>(0.001)   | 0.041***<br>(0.001)  | 0.024***<br>(0.001)   | 0.024***<br>(0.001)  |
| \$50,000–75,000                 |  | 0.029***<br>(0.001)   | 0.029***<br>(0.001)  | 0.016***<br>(0.000)   | 0.016***<br>(0.000)  |
| \$75,000+                       |  | 0.018***<br>(0.001)   | 0.017***<br>(0.001)  | 0.009***<br>(0.000)   | 0.009***<br>(0.000)  |
| Self-employed                   |  | -0.001<br>(0.001)     | -0.001<br>(0.001)    | 0.002***<br>(0.001)   | 0.002***<br>(0.001)  |

|                                  | MDD ( $\geq 14$ days) |                      | MDD ( $\geq 21$ days) |                      |
|----------------------------------|-----------------------|----------------------|-----------------------|----------------------|
|                                  | (1)                   | (2)                  | (3)                   | (4)                  |
| No job                           | 0.186***<br>(0.003)   | 0.186***<br>(0.003)  | 0.132***<br>(0.002)   | 0.132***<br>(0.002)  |
| Homemaker                        | 0.006***<br>(0.002)   | 0.006***<br>(0.002)  | 0.006***<br>(0.001)   | 0.006***<br>(0.001)  |
| Student                          | 0.015***<br>(0.002)   | 0.015***<br>(0.002)  | -0.003**<br>(0.001)   | -0.003**<br>(0.001)  |
| Retired                          | 0.012***<br>(0.001)   | 0.012***<br>(0.001)  | 0.011***<br>(0.001)   | 0.011***<br>(0.001)  |
| Percent Black                    | 0.135*<br>(0.070)     | 0.152**<br>(0.073)   | 0.052<br>(0.044)      | 0.059<br>(0.046)     |
| Percent Hispanic                 | -0.256***<br>(0.079)  | -0.231***<br>(0.081) | -0.175***<br>(0.057)  | -0.160***<br>(0.058) |
| Percent no access to internet    | 0.02<br>(0.024)       | 0.019<br>(0.025)     | 0.003<br>(0.012)      | 0.002<br>(0.013)     |
| Percent unemployed               | -0.003<br>(0.047)     | -0.014<br>(0.048)    | 0.003<br>(0.029)      | -0.003<br>(0.030)    |
| Percent uninsured                | -0.002<br>(0.033)     | 0.001<br>(0.033)     | -0.01<br>(0.020)      | -0.007<br>(0.021)    |
| Telehealth payment parity        | -0.001<br>(0.001)     | -0.001<br>(0.001)    | -0.001**<br>(0.001)   | -0.002*<br>(0.001)   |
| Emergency expansions             |                       |                      |                       |                      |
| Out-of-state license endorsement | 0<br>(0.001)          | 0<br>(0.001)         | 0<br>(0.001)          | 0<br>(0.001)         |
| Telehealth                       | 0.002<br>(0.003)      | 0.001<br>(0.002)     | 0.001<br>(0.002)      | 0<br>(0.002)         |
| Observations                     | 5,683,627             | 5,683,627            | 5,683,627             | 5,683,627            |
| R <sup>2</sup>                   | 0.081                 | 0.081                | 0.058                 | 0.058                |
| Mean dependent variable          | 0.115                 | 0.115                | 0.066                 | 0.066                |

*Panel B: Ever Diagnosed with Depression*

|                    |                    |                    |                      |                      |
|--------------------|--------------------|--------------------|----------------------|----------------------|
| IMLC               | 0.007**<br>(0.003) |                    | 0.002<br>(0.002)     |                      |
| PSYPACT            | 0.002<br>(0.004)   |                    | 0.002<br>(0.003)     |                      |
| Either compact     |                    | 0.007**<br>(0.003) |                      | 0.002<br>(0.002)     |
| Male               | -0.003*<br>(0.002) | -0.003*<br>(0.002) | 0.011***<br>(0.001)  | 0.011***<br>(0.001)  |
| Race and ethnicity |                    |                    |                      |                      |
| Black              | -0.003<br>(0.003)  | -0.003<br>(0.003)  | -0.011***<br>(0.004) | -0.011***<br>(0.004) |
| Other              | 0.032***           | 0.032***           | 0.029***             | 0.029***             |

|                                 | MDD ( $\geq 14$ days) |           | MDD ( $\geq 21$ days) |           |
|---------------------------------|-----------------------|-----------|-----------------------|-----------|
|                                 | (1)                   | (2)       | (3)                   | (4)       |
|                                 | (0.003)               | (0.003)   | (0.003)               | (0.003)   |
| Hispanic                        | -0.017***             | -0.017*** | -0.015***             | -0.015*** |
|                                 | (0.004)               | (0.004)   | (0.003)               | (0.003)   |
| Uninsured                       | 0.024***              | 0.024***  | 0.028***              | 0.028***  |
|                                 | (0.003)               | (0.003)   | (0.003)               | (0.003)   |
| Married                         | -0.012***             | -0.012*** | -0.001                | -0.001    |
|                                 | (0.003)               | (0.003)   | (0.002)               | (0.002)   |
| Divorced                        | 0.025***              | 0.025***  | 0.028***              | 0.028***  |
|                                 | (0.002)               | (0.002)   | (0.001)               | (0.001)   |
| Widowed                         | 0.006**               | 0.006**   | 0.018***              | 0.018***  |
|                                 | (0.003)               | (0.002)   | (0.002)               | (0.002)   |
| Age                             |                       |           |                       |           |
| 35–44                           | -0.028***             | -0.028*** | 0                     | 0         |
|                                 | (0.002)               | (0.002)   | (0.002)               | (0.002)   |
| 45–54                           | -0.045***             | -0.045*** | -0.005***             | -0.005*** |
|                                 | (0.003)               | (0.003)   | (0.002)               | (0.002)   |
| 55–64                           | -0.093***             | -0.093*** | -0.035***             | -0.035*** |
|                                 | (0.003)               | (0.003)   | (0.002)               | (0.002)   |
| 65+                             | -0.157***             | -0.157*** | -0.077***             | -0.077*** |
|                                 | (0.003)               | (0.003)   | (0.003)               | (0.003)   |
| High school or less             | 0.002                 | 0.002     | -0.007                | -0.007    |
|                                 | (0.009)               | (0.009)   | (0.010)               | (0.010)   |
| Some college/Associate's degree | -0.007                | -0.007    | -0.017                | -0.017    |
|                                 | (0.009)               | (0.009)   | (0.010)               | (0.010)   |
| Bachelor's degree or higher     | -0.063***             | -0.063*** | -0.056***             | -0.056*** |
|                                 | (0.009)               | (0.009)   | (0.010)               | (0.010)   |
| Income                          |                       |           |                       |           |
| \$15,000–25,000                 | 0.153***              | 0.153***  | 0.117***              | 0.117***  |
|                                 | (0.003)               | (0.003)   | (0.002)               | (0.002)   |
| \$25,000–35,000                 | 0.120***              | 0.120***  | 0.080***              | 0.080***  |
|                                 | (0.002)               | (0.002)   | (0.002)               | (0.002)   |
| \$35,000–50,000                 | 0.087***              | 0.087***  | 0.057***              | 0.057***  |
|                                 | (0.002)               | (0.002)   | (0.002)               | (0.002)   |
| \$50,000–75,000                 | 0.068***              | 0.068***  | 0.041***              | 0.041***  |
|                                 | (0.002)               | (0.002)   | (0.002)               | (0.002)   |
| \$75,000+                       | 0.043***              | 0.043***  | 0.025***              | 0.025***  |
|                                 | (0.002)               | (0.002)   | (0.001)               | (0.001)   |
| Self-employed                   | -0.004**              | -0.004**  | 0.004**               | 0.004**   |
|                                 | (0.002)               | (0.002)   | (0.002)               | (0.002)   |
| No job                          | 0.210***              | 0.210***  | 0.170***              | 0.170***  |
|                                 | (0.002)               | (0.002)   | (0.002)               | (0.002)   |
| Homemaker                       | 0.025***              | 0.025***  | 0.021***              | 0.021***  |

|                                  | MDD ( $\geq 14$ days) |                     | MDD ( $\geq 21$ days) |                     |
|----------------------------------|-----------------------|---------------------|-----------------------|---------------------|
|                                  | (1)                   | (2)                 | (3)                   | (4)                 |
| Student                          | (0.004)<br>0.041***   | (0.004)<br>0.041*** | (0.003)<br>0.003      | (0.003)<br>0.003    |
| Retired                          | (0.003)<br>0.032***   | (0.003)<br>0.032*** | (0.003)<br>0.032***   | (0.003)<br>0.032*** |
| Percent Black                    | (0.002)<br>0.394**    | (0.002)<br>0.443**  | (0.002)<br>0.1        | (0.002)<br>0.114    |
| Percent Hispanic                 | (0.169)<br>-0.084     | (0.168)<br>-0.112   | (0.155)<br>-0.045     | (0.155)<br>-0.041   |
| Percent no access to internet    | (0.270)<br>0.072      | (0.269)<br>0.078    | (0.194)<br>0.033      | (0.192)<br>0.034    |
| Percent unemployed               | (0.048)<br>0.111      | (0.047)<br>0.111    | (0.035)<br>0.123      | (0.035)<br>0.12     |
| Percent uninsured                | (0.175)<br>0.094      | (0.174)<br>0.091    | (0.111)<br>0.071      | (0.111)<br>0.071    |
| Telehealth payment parity        | (0.084)<br>-0.001     | (0.081)<br>-0.001   | (0.061)<br>-0.003     | (0.060)<br>-0.003   |
| Emergency expansions             | (0.003)<br>(0.003)    | (0.003)<br>(0.003)  | (0.002)<br>(0.002)    | (0.002)<br>(0.002)  |
| Out-of-state license endorsement | 0.002<br>(0.003)      | 0.002<br>(0.003)    | 0<br>(0.003)          | 0<br>(0.003)        |
| Telehealth                       | 0.005<br>(0.008)      | 0.005<br>(0.008)    | 0.004<br>(0.007)      | 0.004<br>(0.007)    |
| Observations                     | 1,077,112             | 1,077,112           | 1,077,112             | 1,077,112           |
| R <sup>2</sup>                   | 0.097                 | 0.097               | 0.075                 | 0.075               |
| Mean dependent variable          | 0.360                 | 0.360               | 0.220                 | 0.220               |

*Notes:* Standard errors are clustered by state. All models include state and year fixed effects. Models additionally control for temporary COVID-19 emergency expansions permitting out-of-state license endorsement and telehealth practice. \*\*\* $p < 0.01$ , \*\* $p < 0.05$ , \* $p < 0.10$ .

**Table A8:** State-level Mental Health and Compact Participation

|                                  | Suicide Rate          |                        | SSRI Prescribing Rate |
|----------------------------------|-----------------------|------------------------|-----------------------|
|                                  | (1)                   | (2)                    | (3)                   |
| IMLC                             | 0.493**<br>(0.213)    |                        | 0.000<br>(0.000)      |
| PSYPACT                          | -0.370<br>(0.222)     |                        |                       |
| Either compact                   |                       | 0.343<br>(0.211)       |                       |
| Percent Black                    | 6.581<br>(15.040)     | 10.217<br>(15.202)     | 0.025*<br>(0.013)     |
| Percent Hispanic                 | -49.702**<br>(18.951) | -54.581***<br>(18.444) | -0.007<br>(0.035)     |
| Percent no access to internet    | -0.115<br>(4.326)     | 0.016<br>(4.271)       | -0.010<br>(0.012)     |
| Percent unemployed               | 20.958**<br>(8.826)   | 21.732**<br>(8.683)    | 0.006<br>(0.006)      |
| Percent uninsured                | -4.791<br>(8.883)     | -5.575<br>(8.868)      | 0.013<br>(0.009)      |
| Telehealth payment parity        | -0.300<br>(0.246)     | -0.308<br>(0.252)      | 0.000<br>(0.000)      |
| Emergency expansions             |                       |                        |                       |
| Out-of-state license endorsement | 0.035<br>(0.242)      | -0.015<br>(0.242)      | 0.000<br>(0.000)      |
| Telehealth                       | 0.499<br>(0.336)      | 0.586<br>(0.352)       | 0.000<br>(0.000)      |
| Observations                     | 663                   | 663                    | 510                   |
| R <sup>2</sup>                   | 0.955                 | 0.954                  | 0.987                 |
| Mean dependent variable          | 15.62                 | 15.62                  | 0.031                 |

*Notes:* Standard errors are clustered by state. All models include state and year fixed effects. Models additionally control for temporary COVID-19 emergency expansions permitting out-of-state license endorsement and telehealth practice. \*\*\* $p < 0.01$ , \*\* $p < 0.05$ , \* $p < 0.10$ .

**Table A9:** Interstate Licensure Compacts and Provider Licensure: Ordered Logit Estimates

|              | 1 State<br>(1)         | 2 States<br>(2)       | 3 States<br>(3)       | 4+ States<br>(4)      |
|--------------|------------------------|-----------------------|-----------------------|-----------------------|
| Primary care | -0.0103***<br>(0.0007) | 0.0095***<br>(0.0006) | 0.0007***<br>(0.0000) | 0.0002***<br>(0.0000) |
| Sample share | 0.950                  | 0.046                 | 0.003                 | 0.001                 |
| Psychiatry   | -0.0097***<br>(0.0022) | 0.0087***<br>(0.0020) | 0.0007***<br>(0.0002) | 0.0003***<br>(0.0001) |
| Sample share | 0.946                  | 0.048                 | 0.005                 | 0.001                 |
| Psychology   | -0.0023**<br>(0.0011)  | 0.0022**<br>(0.0010)  | 0.0001**<br>(0.0001)  | 0.0000<br>(0.0000)    |
| Sample share | 0.983                  | 0.016                 | 0.001                 | 0.000                 |

*Notes:* The table reports average marginal effects from ordered logit models of the number of states in which a provider holds an active license. The dependent variable categorizes providers as licensed in one, two, three, or four or more states. Estimates for primary care physicians and psychiatrists represent the discrete change in the predicted probability of being in the indicated licensure category associated with IMLC participation. Estimates for psychologists represent the corresponding change associated with PSYPACT participation. Sample-share rows report the proportion of provider-year observations in each category for the corresponding estimation sample. Models exclude 2017 and include year fixed effects. Standard errors, clustered at the provider level, are reported in parentheses. Specialty categories are not necessarily mutually exclusive. \*\*\* $p < 0.01$ , \*\* $p < 0.05$ , \* $p < 0.10$ .